

Broadcast Technology UG1

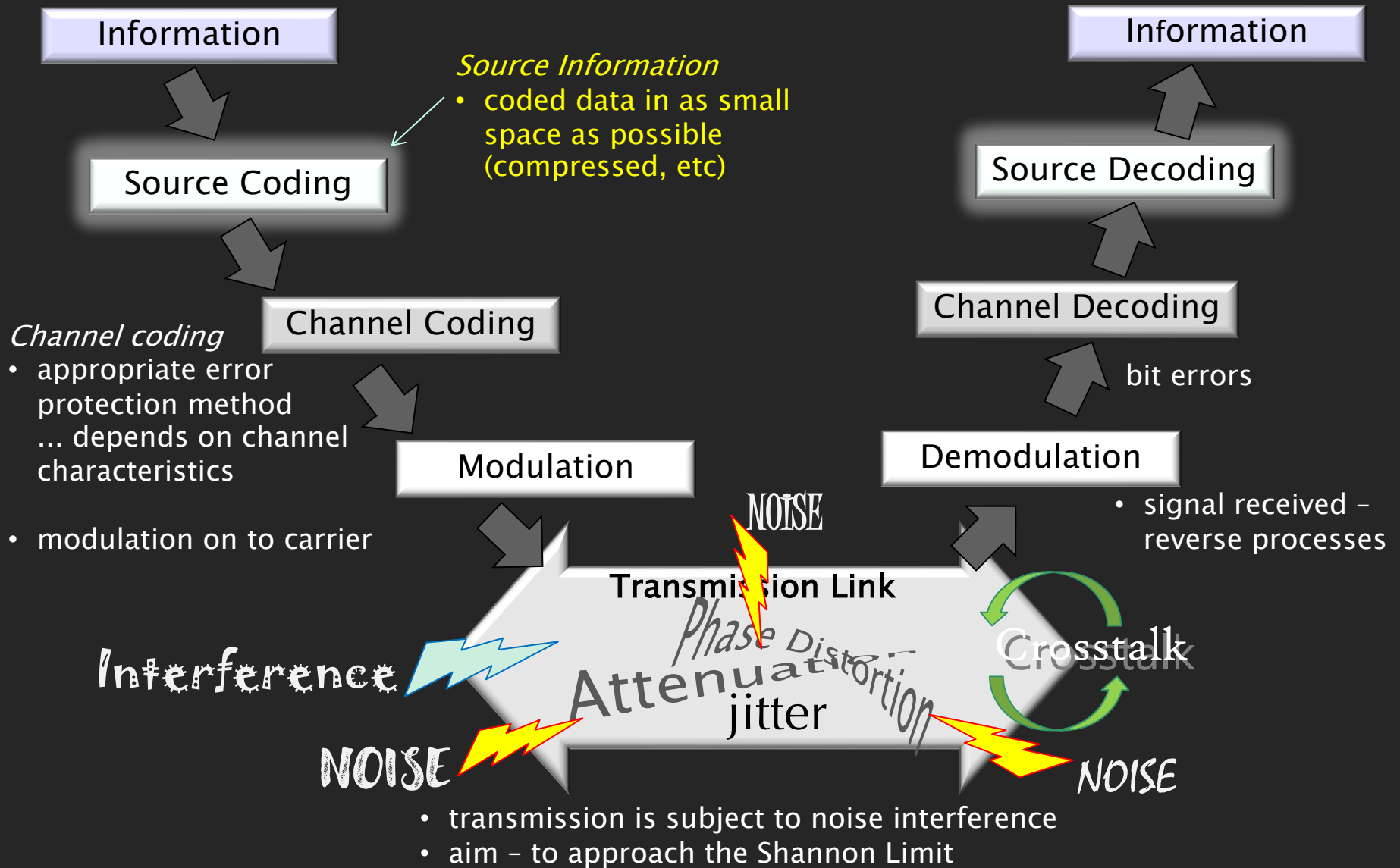
Transmission Media

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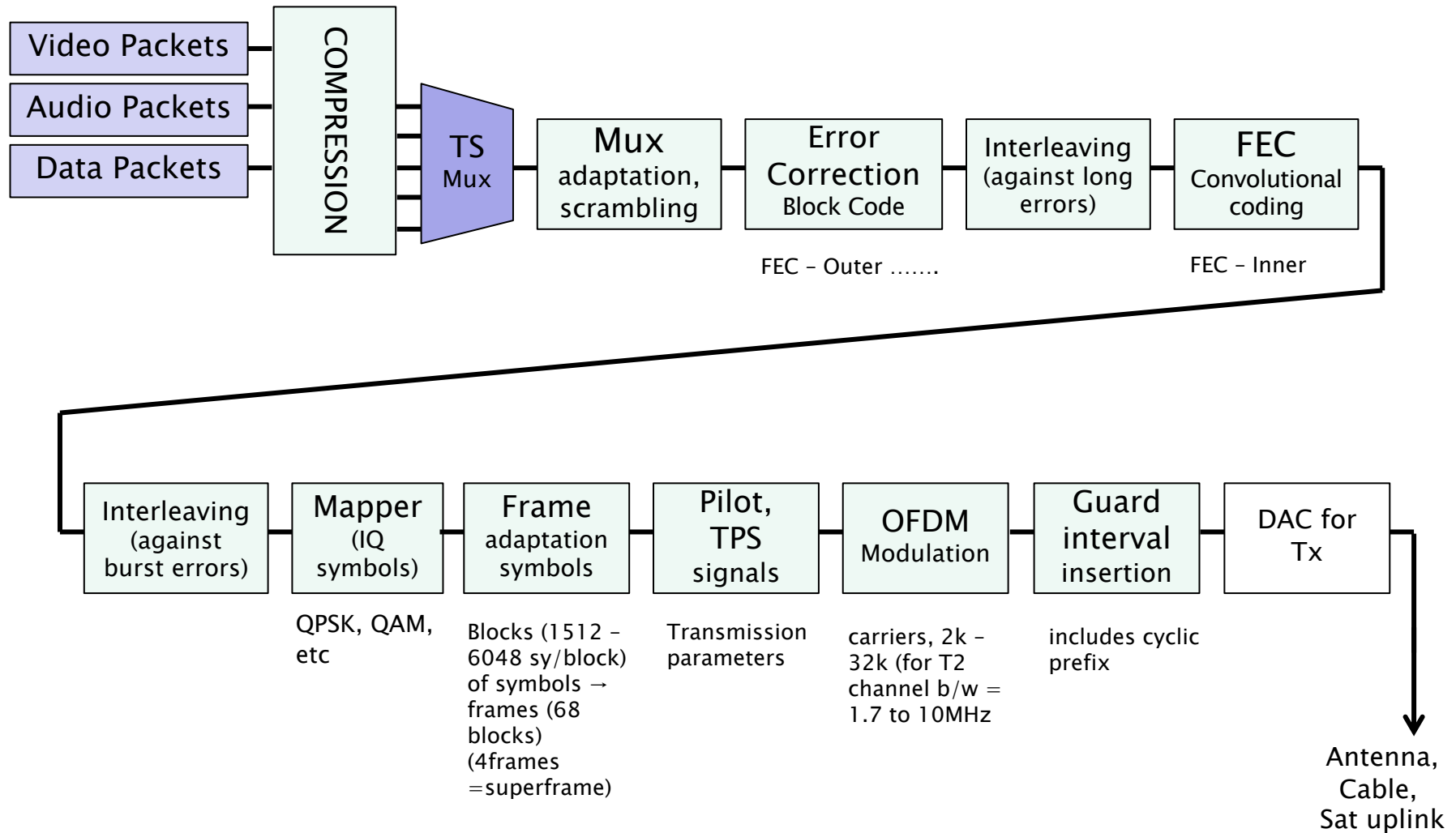
The Context ...

Information Transmission, Signal Encoding



DVB Transmission Playout System

(Based on DVB-T, but -C, -S and -H are similar)



Back to Basics

Transmission Media

Transmission Media

- Types of Transmission Media
- Multiplexing – FDM, TDM, Statistical
 - * *introduction only – will be covered in depth later on*
- Cables
 - Twisted Pair
 - Coaxial
- Fibre Optic
 - Wave Division Multiplexing
- Radio
 - Spectrum
 - Transmitter
 - EM Wave, Radio Waves
 - Propagation through lower and upper atmosphere
 - Broadcast Transmitter Technologies (valves)

Types of Transmission Media

“Guided”: The waves are guided along a physical path

“Unguided”: The medium transmits the waves but does not guide them

Types of Transmission Connection

Direct Link:

- The signal is propagated directly from transmitter to receiver (perhaps through amplifiers/repeaters)

Point-to-Point:

- A guided medium that provides a direct link between two and only two devices

Multipoint:

- A guided medium that is shared by more than two devices

Twisted Pair Cable

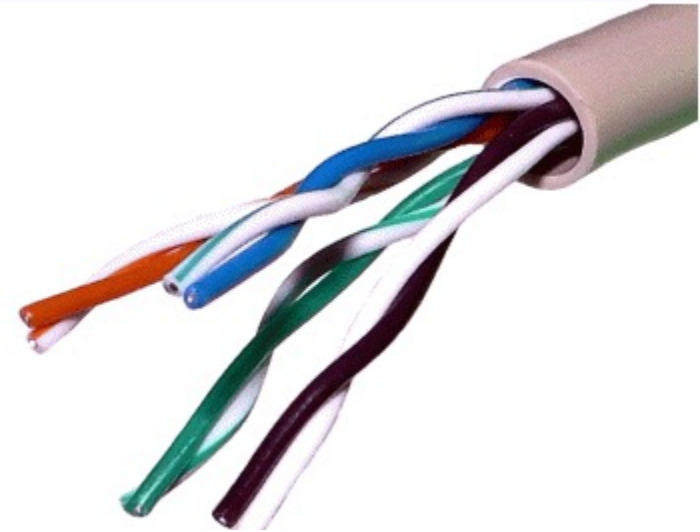
- Two insulated copper wires that are twisted together (which results in cancellation of induced interference; cancellation of radiated interference, hence reduces radiation)
- The individual pairs can be shielded (STP) or unshielded (UTP)
- The cable can also be shielded (S/STP)

Categories (CAT ...)

- | | |
|---|---|
| 1 | – primarily for voice |
| 2 | – 4Mbps |
| 3 | – 16MHz bandwidth; 10Mb/s |
| 4 | – 20MHz bandwidth; 16Mb/s |
| 5 | – 100MHz bandwidth; 100Mb/s (4 pairs/cable) |
| 6 | – 400MHz bandwidth; 1Gb/s (4 pairs/cable) |
| 7 | – 600MHz bandwidth (1GHz for Augmented); 10Gb/s |
| 8 | – 1.2GHz bandwidth |

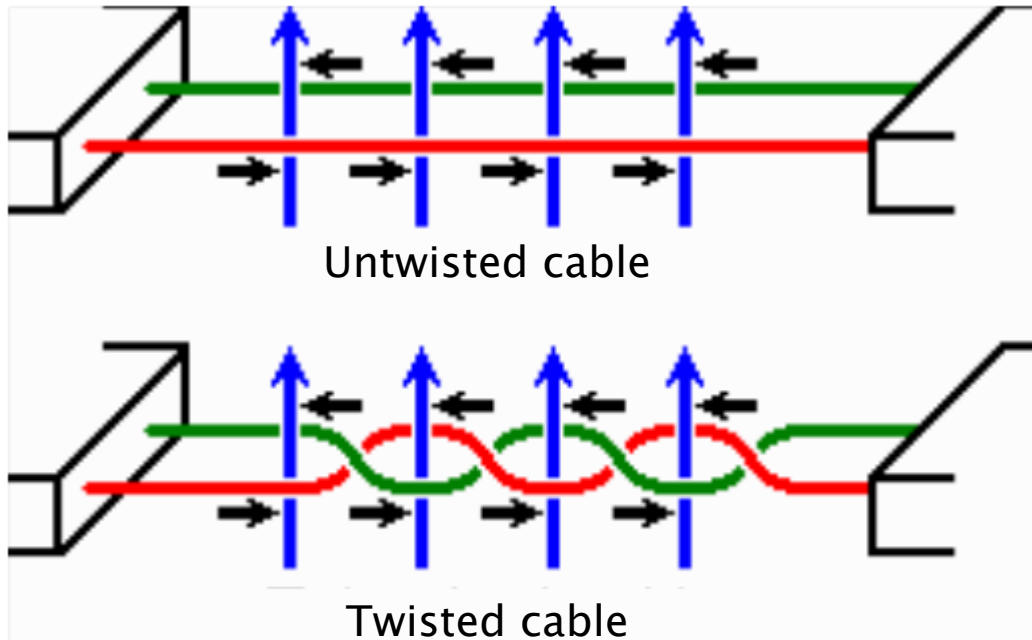
Common Uses:

Voice Telephony, various Digital Subscriber Line (xDSL), LANs



CAT 7 – Screen
Shielded Twisted
Pair (SSTP)

Twisted Pair Cable



Magnetic field



Induced noise current

- Noise generated by magnetic fields from the environment.
- noise current is the result of that magnetic field
- straight cable - all noise current flows same direction
- twisted cable – in alternate twists noise current is the opposite from the current in other twists

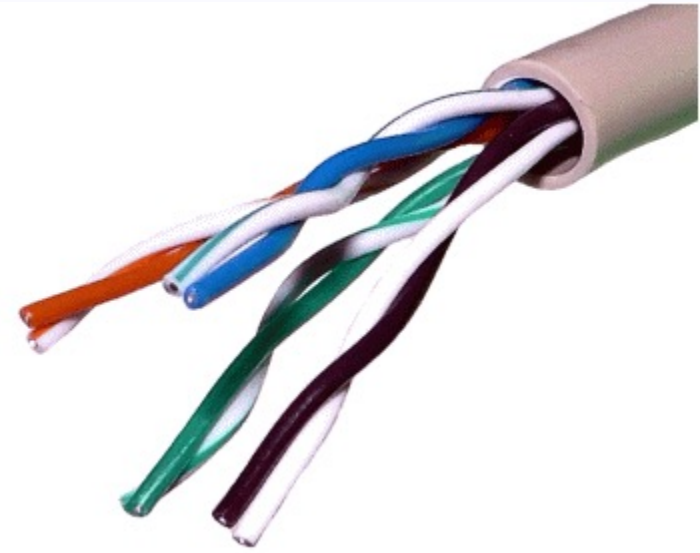
Twisted Pair Cable

Advantages:

- High availability
- Low installation cost

Disadvantages:

- Usable frequency spectrum is small
- Limited data rates
- Limited range
- High susceptibility to electromagnetic and radio frequency interference



ref – Prof. David Bernstein, James Madison University

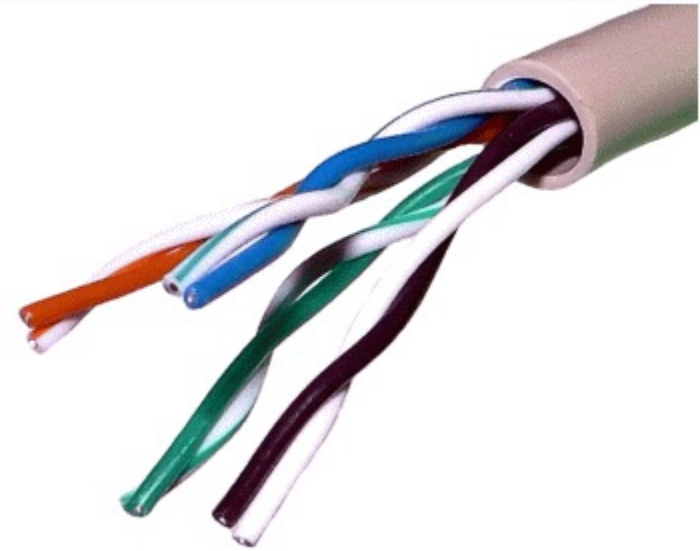
Twisted Pair Cable

In balanced pair operation –

Differential mode transmission –

- differential signals; receiver detects the difference between the two
- noise signals tend to couple to both wires equally
∴ noise produces a common signal in each conductor
– which is cancelled at the receiver

(not effective when the noise source is close to the signal wires;
closer wire will couple with the noise more strongly)



Twist rate

- more twists the better (usually)

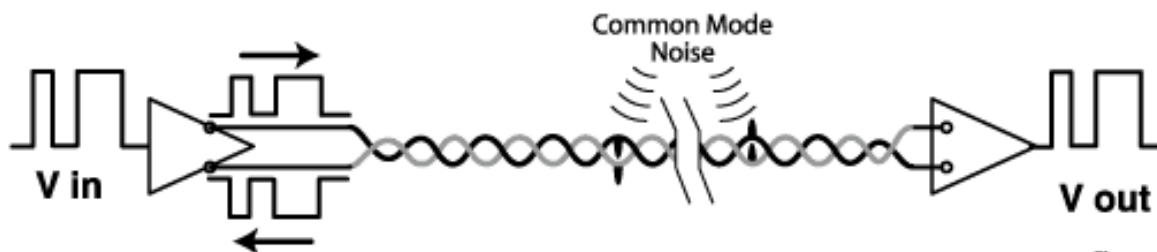


Figure 1.
Common mode noise: The noise is at the same level traveling in the same direction at the same time. The noise will cancel at the receiver.

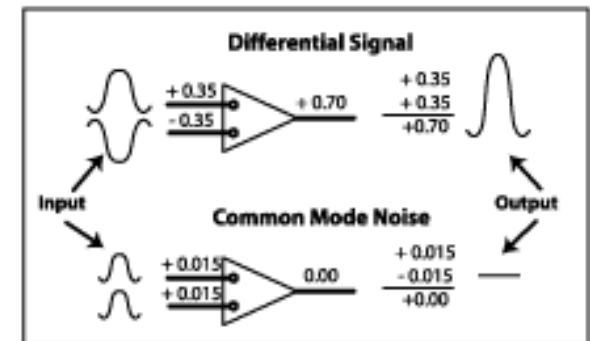
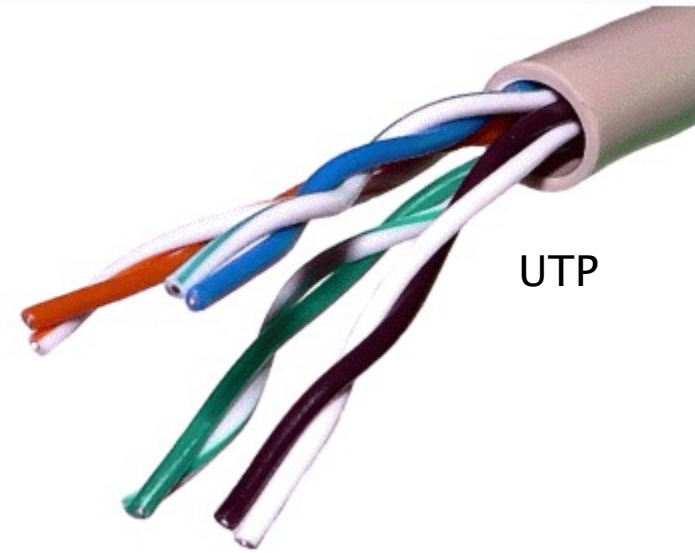
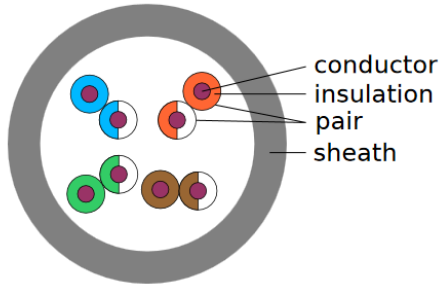


Figure 3.
How common mode noise is cancelled at the receiver.

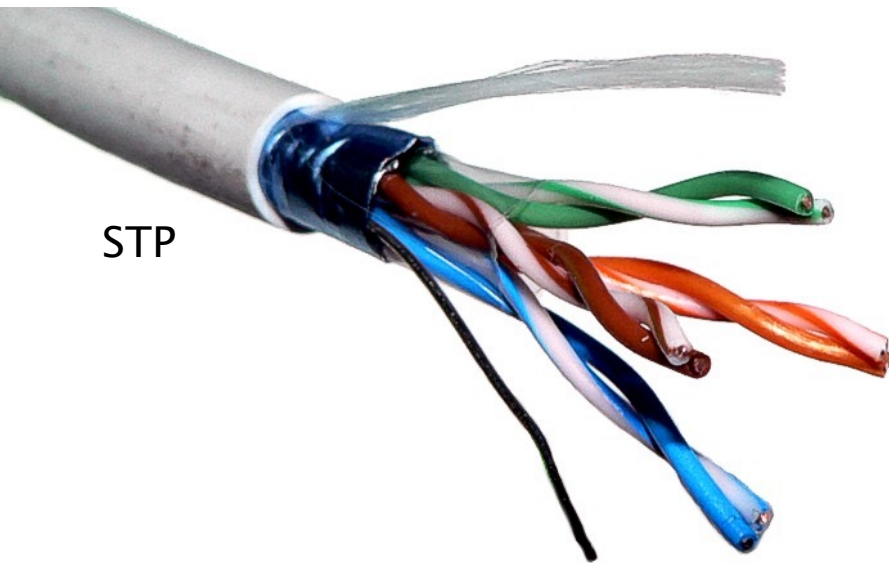
<http://www.extron.com/company/article.aspx?id=utptechnology>

Twisted Pair Cable

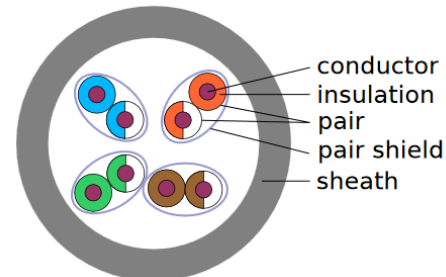
UTP: Unshielded twisted pair



STP: Shielded twisted pair



Twist rate
• more twists the better
(usually)



Twisted Pair Cable

Telecommunication cables pairs in the same cable adjacent for many miles

... one pair can induce crosstalk in another

- each half twist the wire nearest to the noise-source is exchanged
- also reduces electromagnetic radiation from the cable



IEEE Cable Standards for Ethernet on Twisted Pair

10BASE-T

- transmission at 10 Mb/s on unshielded twisted-pair cabling
- defines various aspects of running Ethernet on this cabling.

100BASE-T

- transmission at 100 Mb/s on unshielded twisted-pair cabling
- defines various aspects of running baseband Ethernet on this cabling

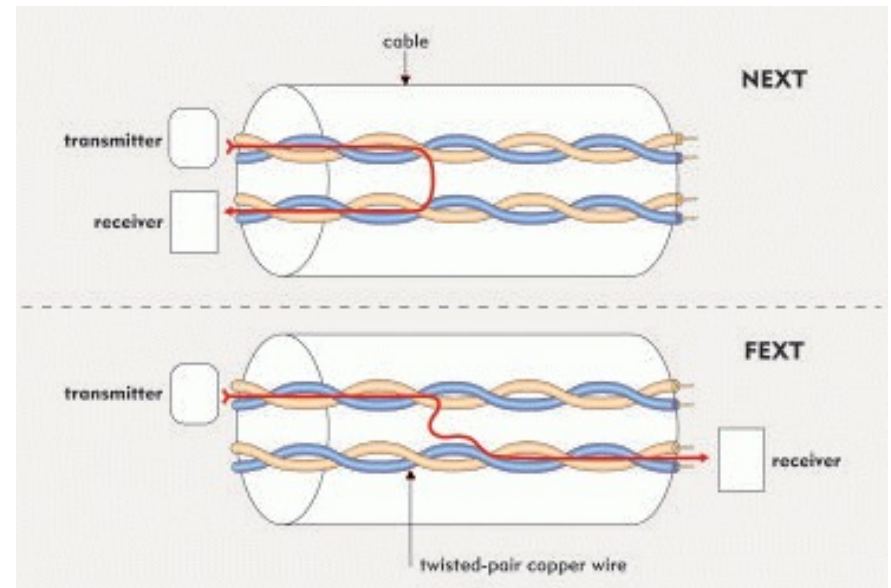
1000BASE-T (gigabit Ethernet) transmission at 1000 Mb/s on unshielded twisted-pair cabling

- defines various aspects of running baseband Ethernet on this cabling

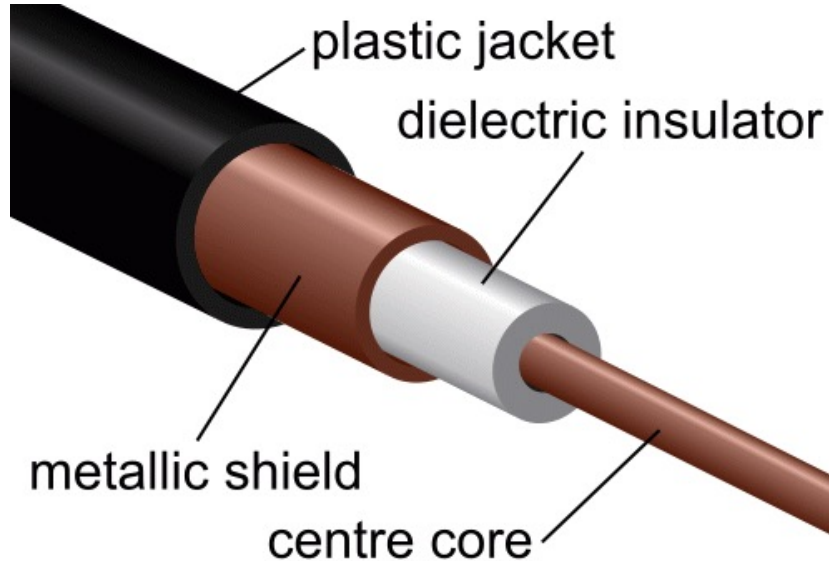
Signal Issues on Twisted Pairs at High Frequencies

- Frequency dependent attenuation and dispersion
 - ... Intersymbol Interference (ISI)
- Coupling between wires in same and adjacent binders
 - ... crosstalk
 - near-end cross-talk = NEXT
 - far-end cross-talk = FEXT
- Bridged taps and gauge transitions
 - ... reflections : pulse distortion (ISI)
- Impulsive noise : high bursts of short duration (10 ms – 1 ms)
- Impedance mismatch between hybrid and line impedance
 - ... reflections (echo)
- RFI interference: AM broadcast & amateur radio
 - ... ingress & egress

All these require a highly adaptive transceiver (at each end)



Coaxial Cable



A (thick) copper wire surrounded by an insulator and an outer conductor

Advantages:

- High capacity (370–1000 times more than twisted pair)
- Low susceptibility to noise and crosstalk (because of shielding)

Disadvantages:

- Noise on the return path
- High installation cost

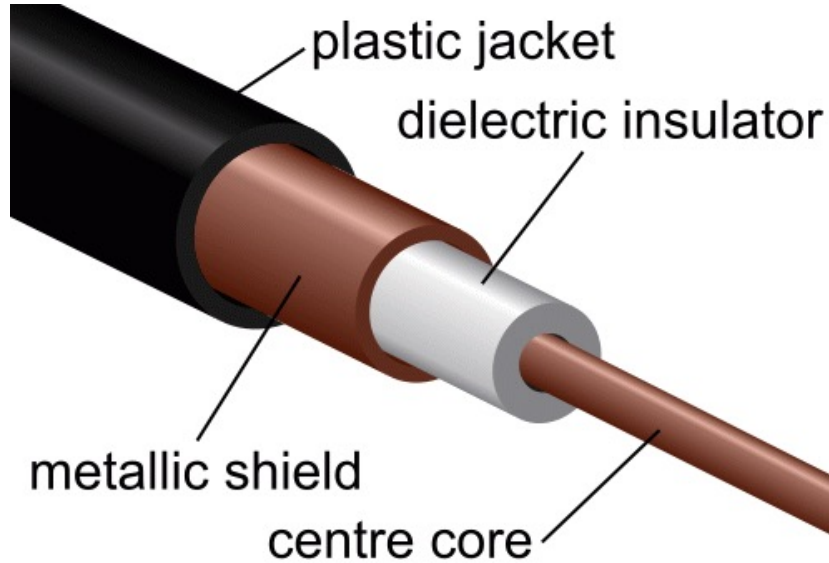
Common Types:

- RG-6 – 1.00mm core, 75 Ω
- RG-59 – 0.81mm core, 75 Ω
- RG-58 – 0.90mm core, 50 Ω
- RG-8 – 2.17mm core, 50 Ω

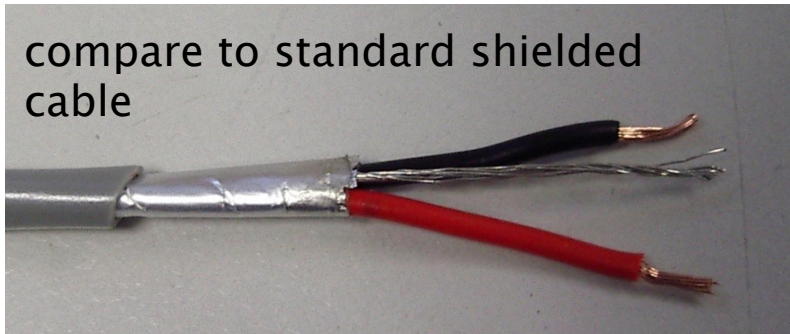
Common Uses:

- Applications for RF signals
- Cable TV (Old: RG-59; New: RG-6)
- Ethernet (RG-58: 10Base2; RG-8: 10Base5)
- connecting radio transmitter feeds

Coaxial Cable

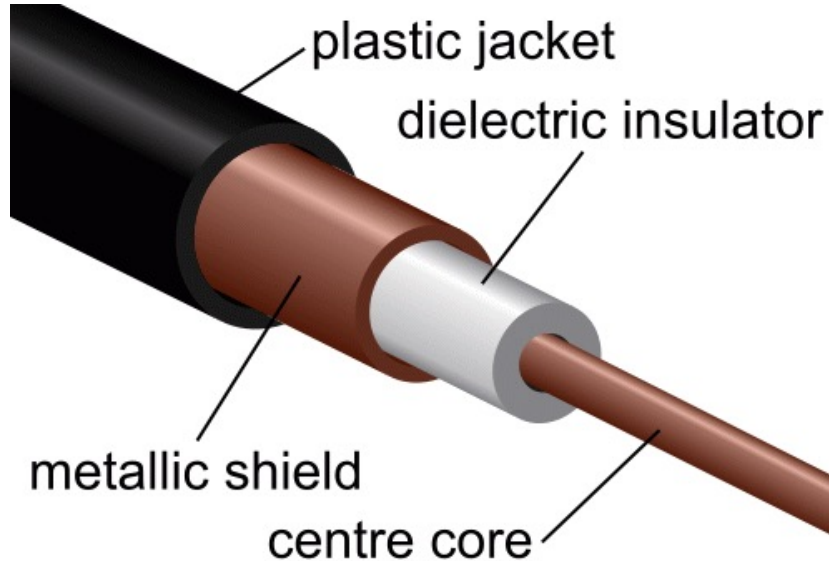


compare to standard shielded cable

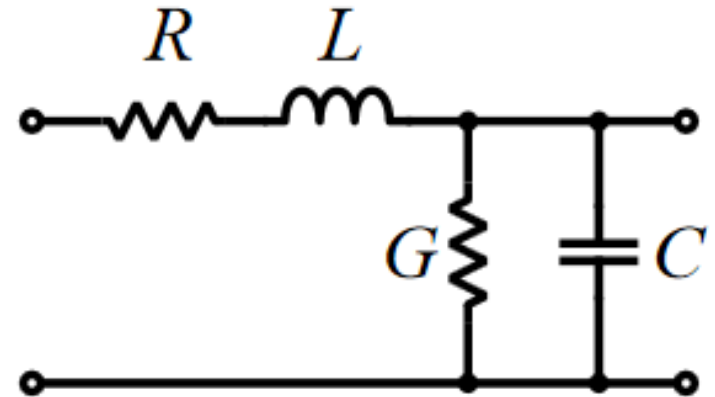


- dimensions of the cable are controlled to give a precise, constant conductor spacing
- needed to function efficiently as a radio frequency transmission line
- electric and magnetic fields are confined to the dielectric with little leakage outside the shield
- electric and magnetic fields outside the cable are largely kept from causing interference to signals inside the cable
- design of cable to control the electrical transmission properties ...

Coaxial Cable



All cables have electrical properties – they are not just short-circuits between equipment



Electrical properties of any cable pair:

R : resistance along the cable

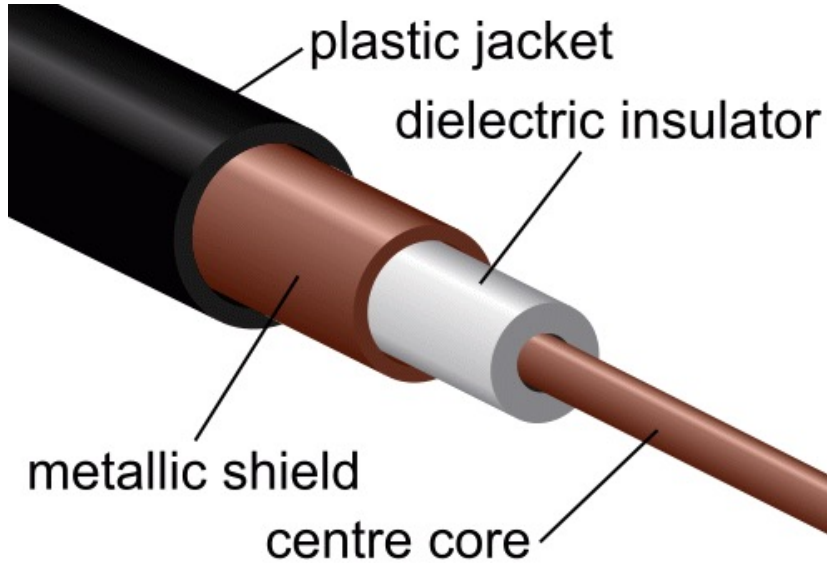
L : signals cause inductance along the length of the cable

G : conductance – a certain amount of leakage, similar to a high resistance between the conductors

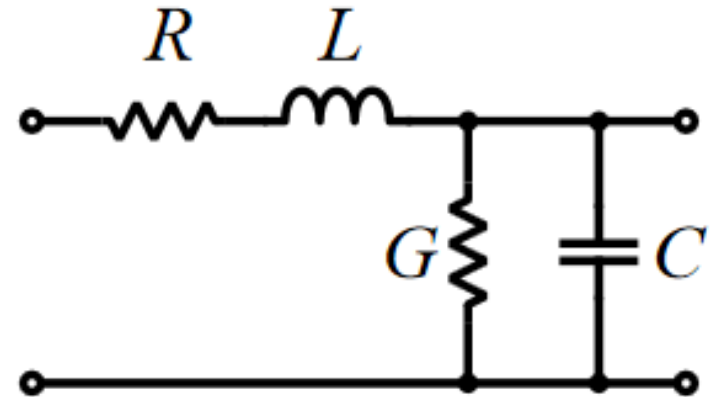
C : capacitance between the two conductors

Coax cable is designed to control these parameters

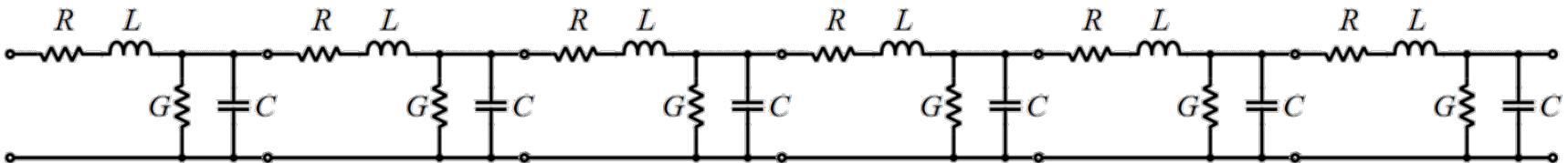
Coaxial Cable



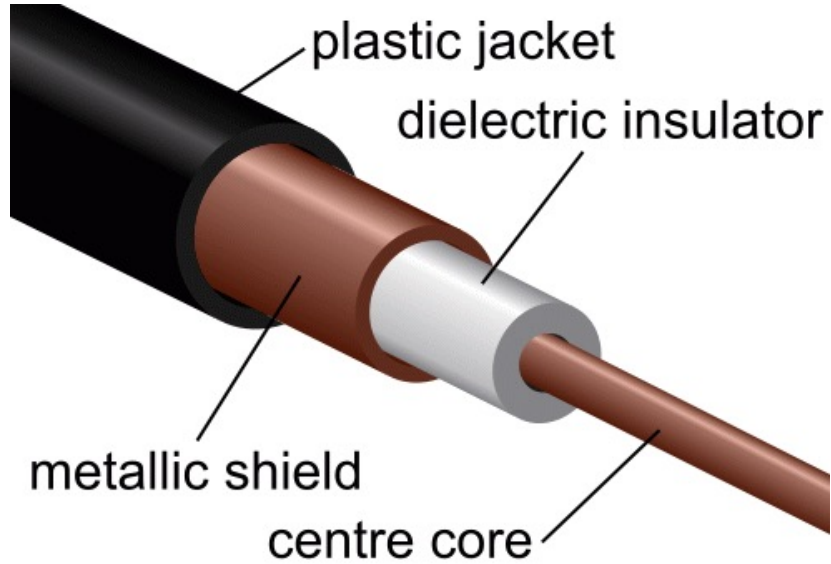
All cables have electrical properties – they are not just short-circuits between equipment



All cables are effectively a (mathematically) complex circuit of distributed impedance sections



Coaxial Cable



Double shielded coax



Coax female socket



Coax male plug

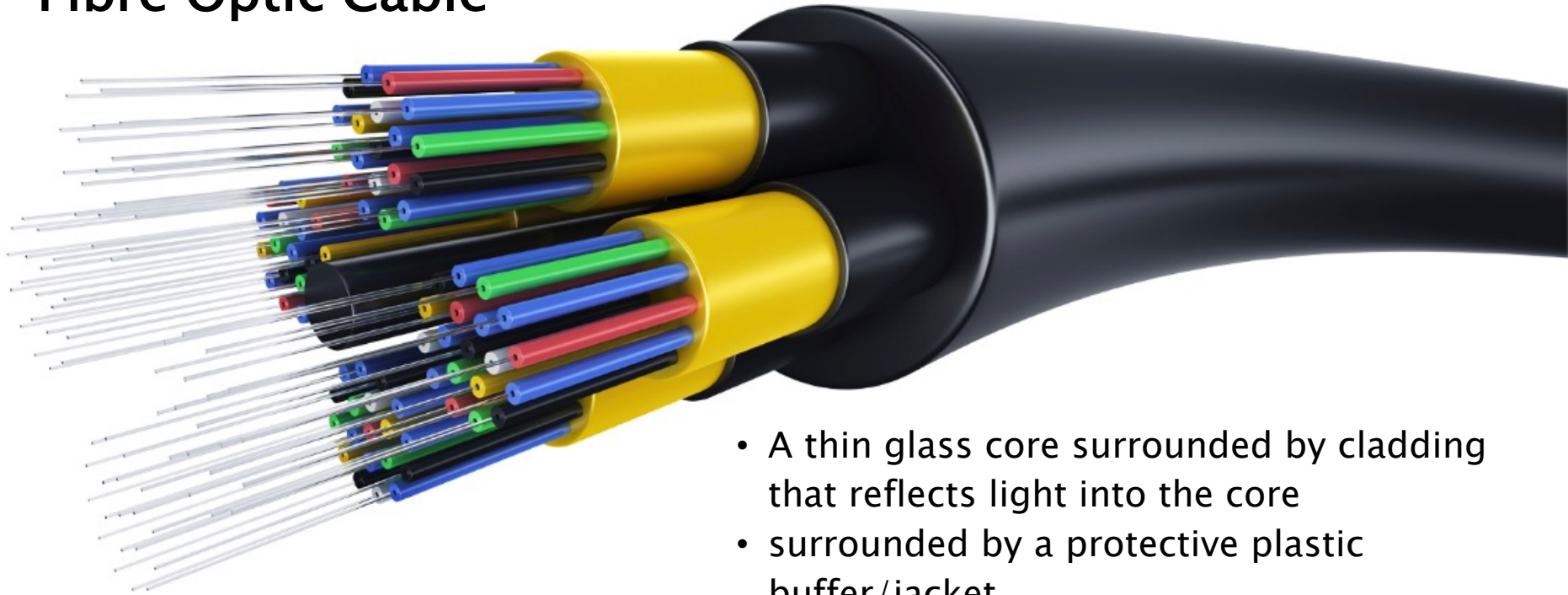
Coaxial Cable



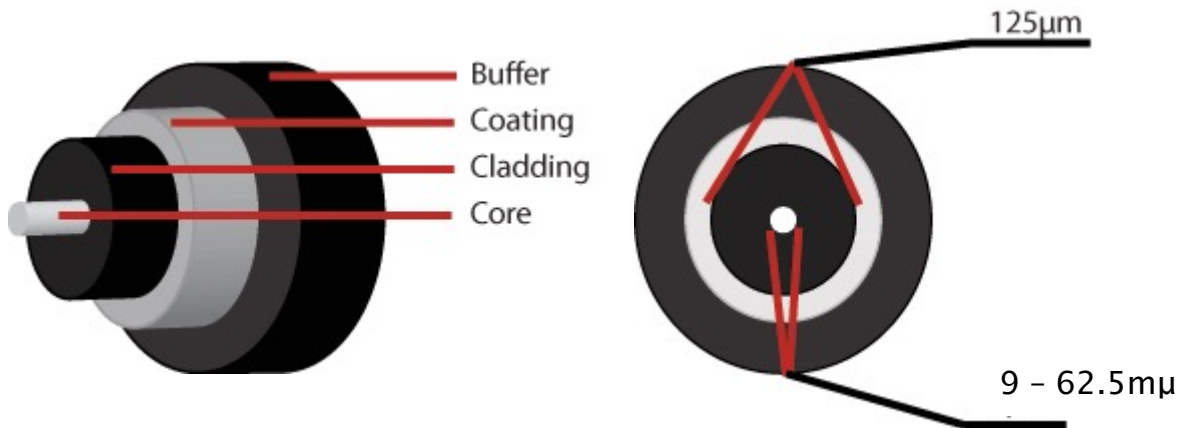
Splitting signals

- simple 2-into-1 can cause impedance imbalance
- splitters should normally terminate each cable and combine in an impedance-balanced circuit, especially at high RF feeds

Fibre Optic Cable



- A thin glass core surrounded by cladding that reflects light into the core
- surrounded by a protective plastic buffer/jacket

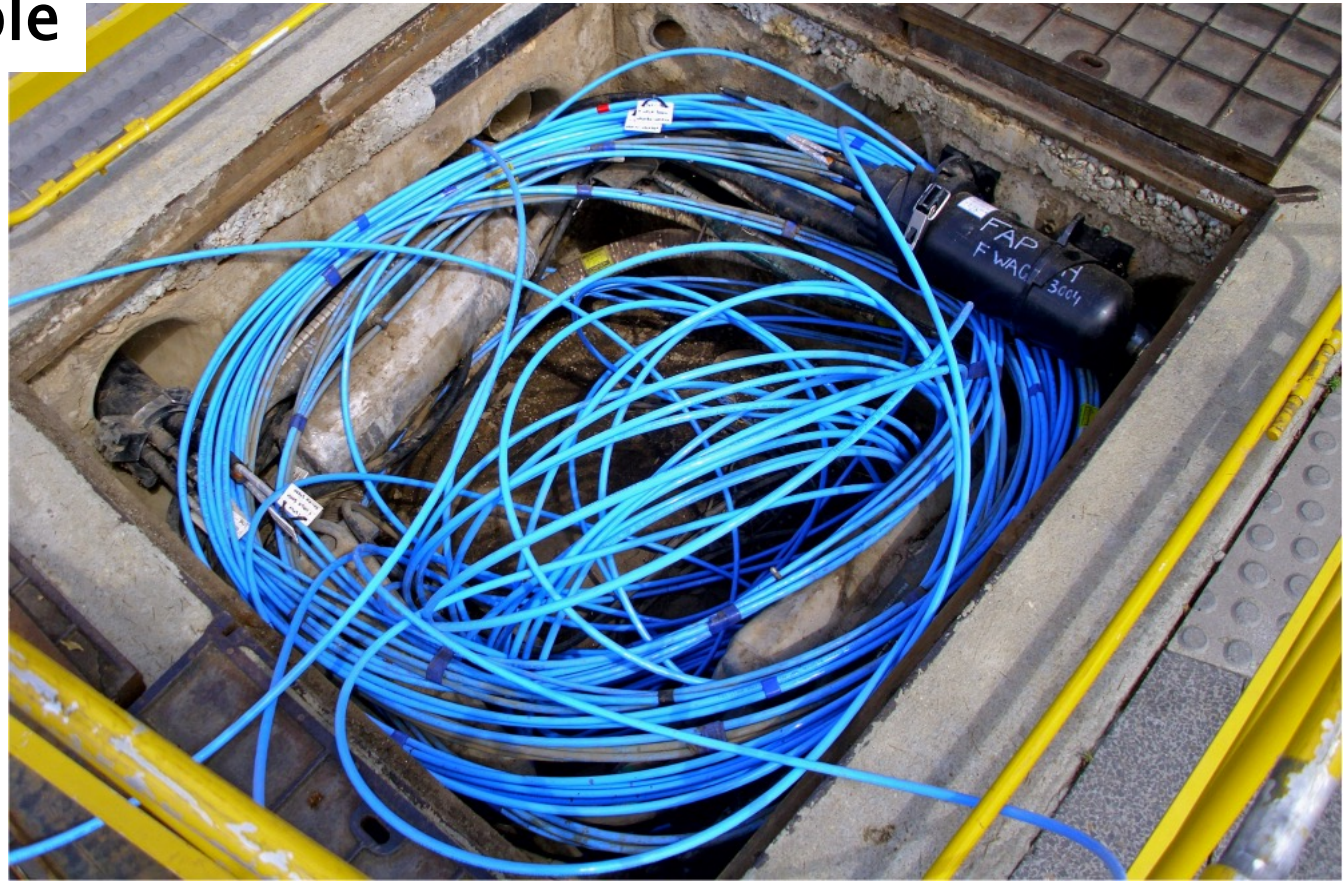


Common Uses:

- Cable TV
- Voice Telephony
- Ethernet

Fibre Optic Cable

Fibre for local
distribution



MicroCore Blown Fibre Cable – for air-blown installation (using existing ducts, e.g. replacing existing cable plant)

Fibre Optic Cable

Common Wavelengths (in the infrared):

- 850nm (i.e., 0.85 μm)
- 1300nm (i.e., 1.30 μm)
- 1550nm (i.e., 1.55 μm)

Advantages:

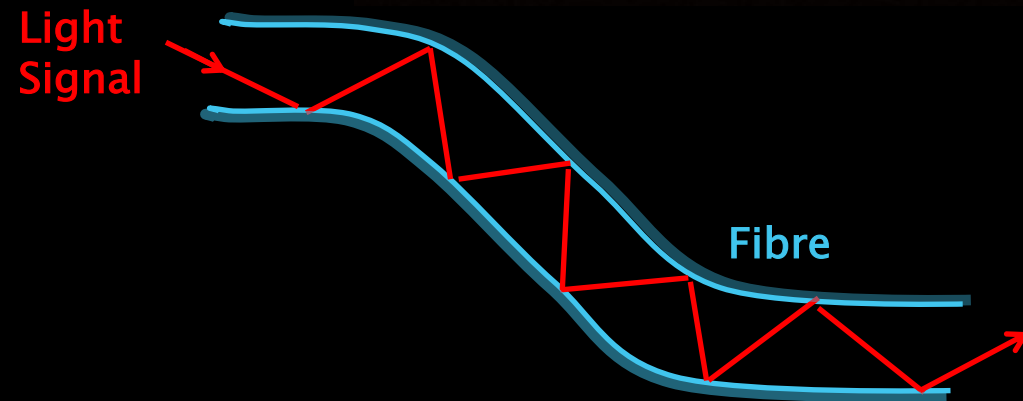
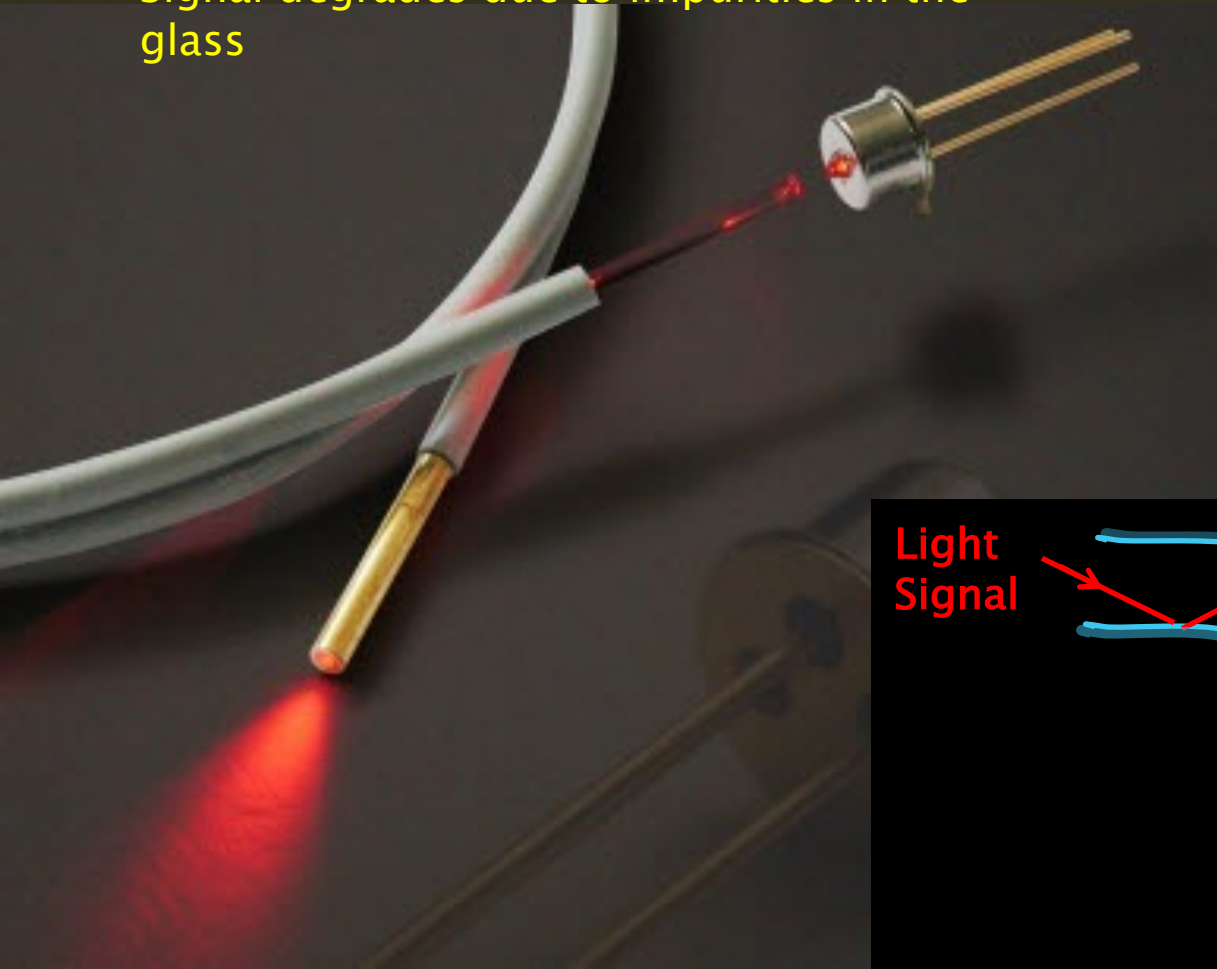
- Low cost
- High capacity
- Thin
- Low interference/degradation

Disadvantages:

- Has a minimum radius for bending
- Each fiber is one-way
- High cost of interfaces

Fibre Optic Cable

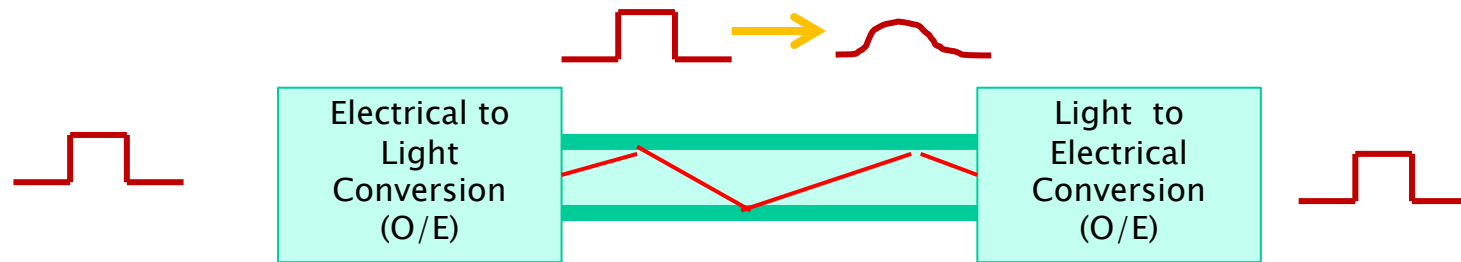
- Light is laser LED;
 - *laser light is a single frequency*
... (i.e. a single wavelength)
- Reflections: Internal reflection of the glass fibre
- Signal degrades due to impurities in the glass



Fibre Optic Cable

Transparent glass or plastic fibres

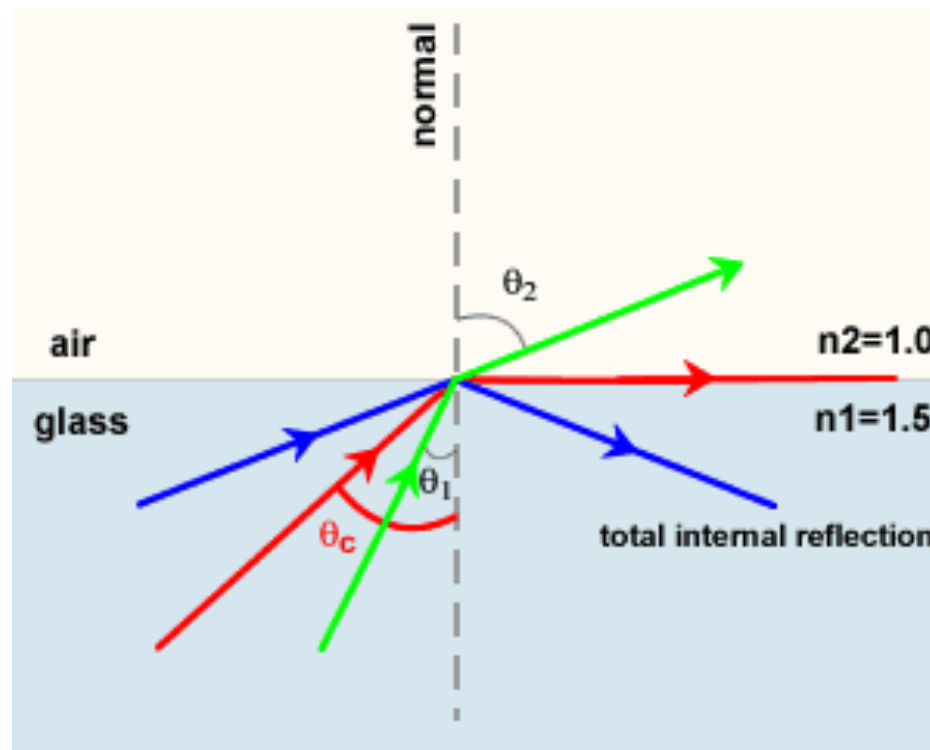
–allow light to be guided from one end to the other with minimal loss



Fibre Optic Cable

Total Internal Reflection

- Light passing from one medium (m_1) to another (m_2) with lower index of refraction (n_2) bends away from the normal line
- At the **critical angle**, defined by $\sin(\theta_c) = \frac{n_2}{n_1}$, the refracted light will not go into the second medium but will travel along the border



Fibre Optic Cable

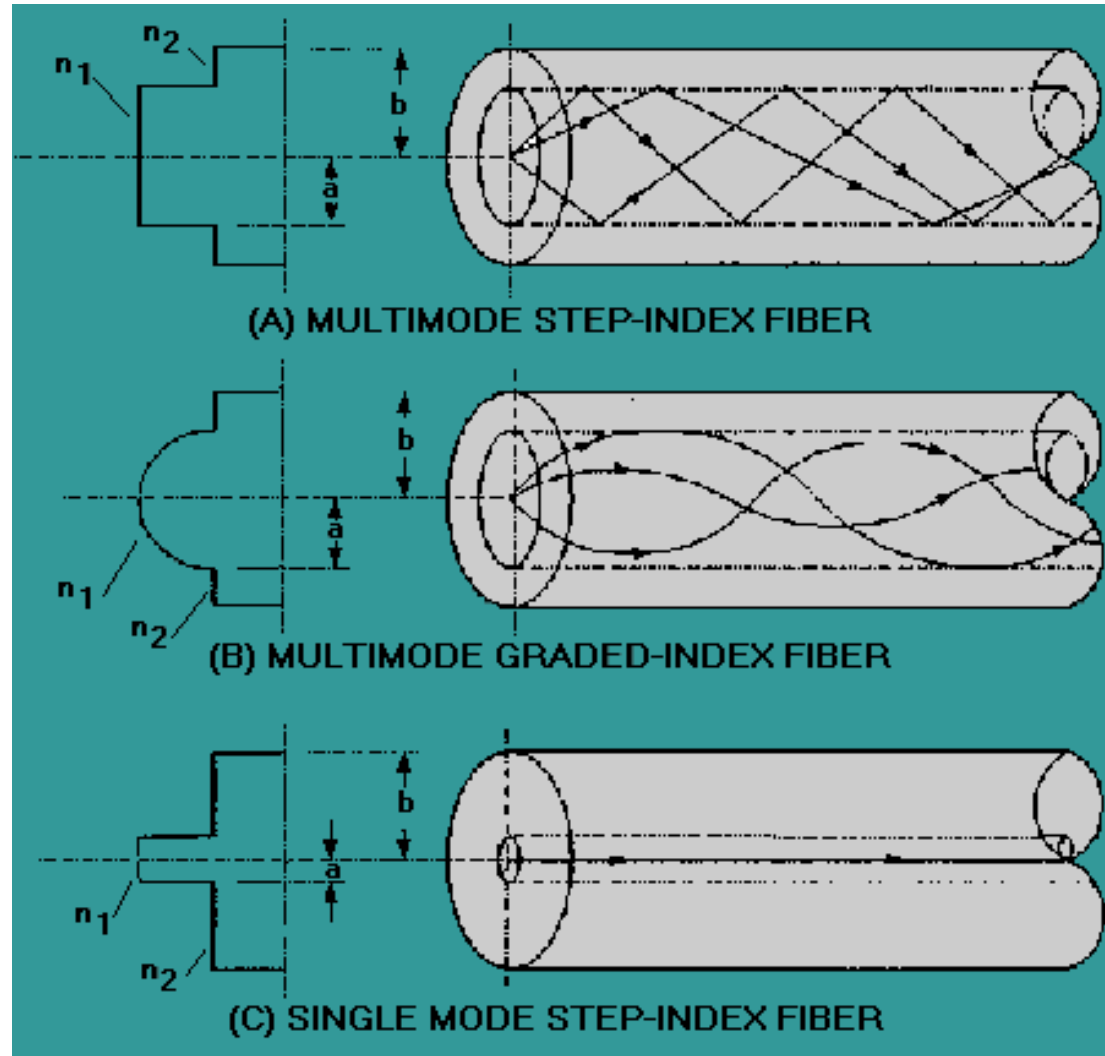
Three types of fibre optic cable commonly used:

Multimode – MMF

- many rays (modes)
- core = 62.5 microns (some 50 microns)
- LED and laser sources 850, 1300nm
- LANs
- graded index ~2GHz b/w step index is rare

Single mode – SMF

- single ray (mode)
- core = 9 microns
- telephony, CATV
- laser @ 1300, 1550 nm
- b/w ~ 100THz



Fibre Optical Multimode, Indexes

Step-Index Multimode Fiber

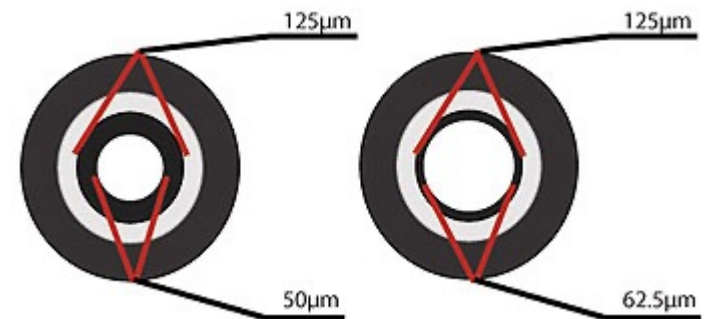
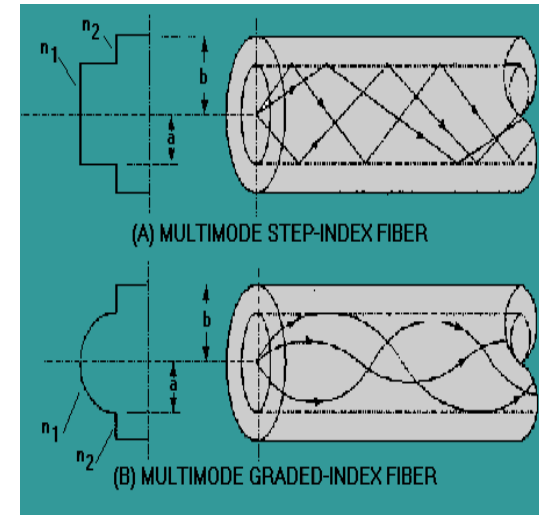
- large core
- some rays direct – others zigzag
 - arrive separately $\therefore$ pulse spreads out
 - leave space ... prevent overlapping
 - limits data rate
- for short(er) distances, <2km, LANs, SANs, data centers etc

Graded-Index Multimode Fiber

- refractive index reduces from the centre
 - centre: higher index
 - rays travel more slowly than those further out
 - rays curve helically
- edge rays – zig-zag
- edge rays and central rays travel (more) equal distances
 $\therefore$ pulse suffers less dispersion

Typically use LEDs as source 850–1310 nm

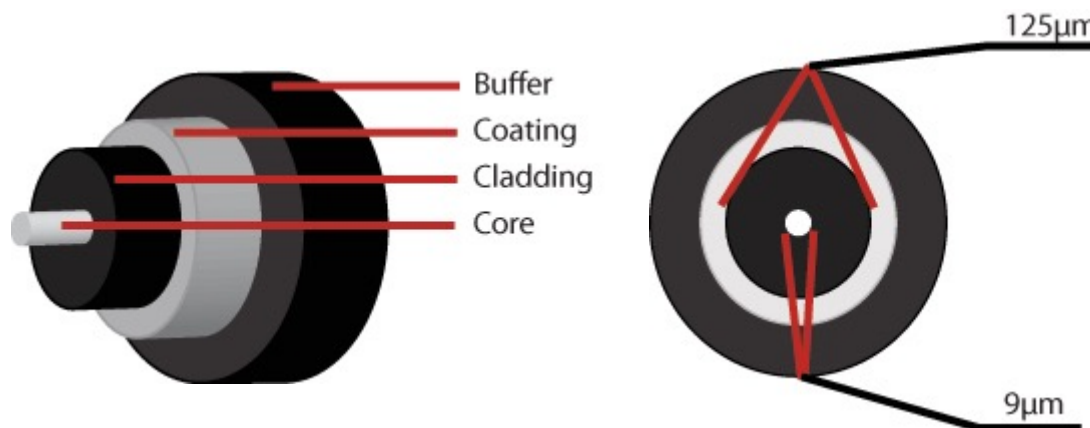
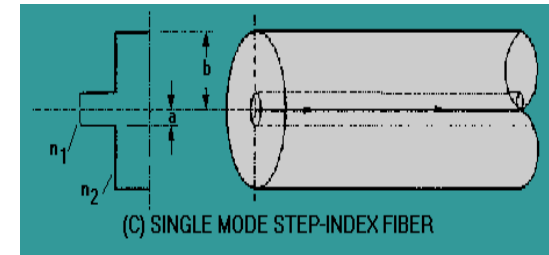
- lower cost
- higher loss
- higher loss



Fibre Optics – Singlemode

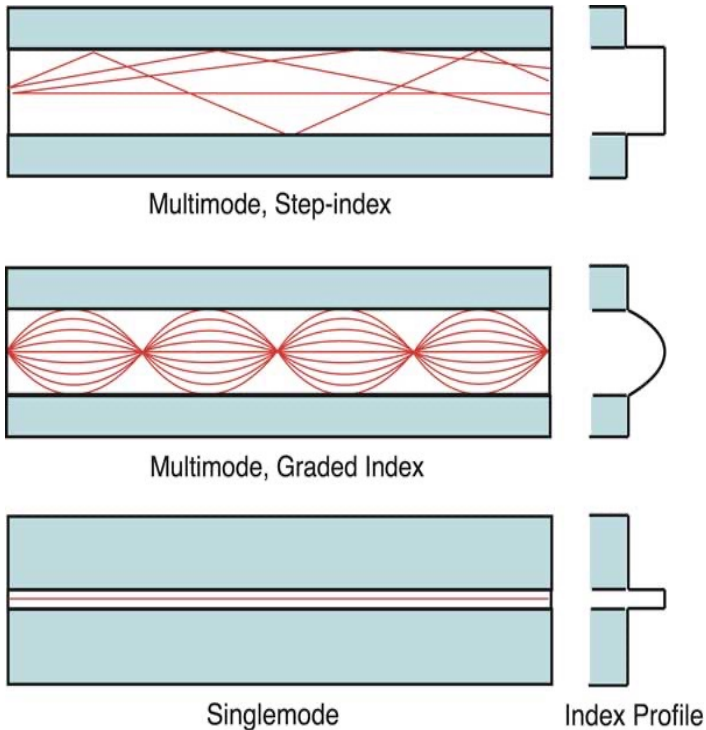
- small diameter core: $9\mu\text{m}$
- allows only one mode of light to propagate
 - number of light reflections created as the light passes through the core decreases
 - lowers attenuation
 - signal can travel further
 - high precision packaging
- typically used in **long distance** $>60\text{km}$; WAN, MAN
 - higher bandwidth runs by Telcos, CATV companies

Typically use lasers as source



Fibre Optics

Summary of Modes and dispersion



Most fibres 125um outside diameter

MMF

- larger core: 50 or 62.5um
- wavelengths: 850 and 1300 nm

SMF

- smaller core ~9um

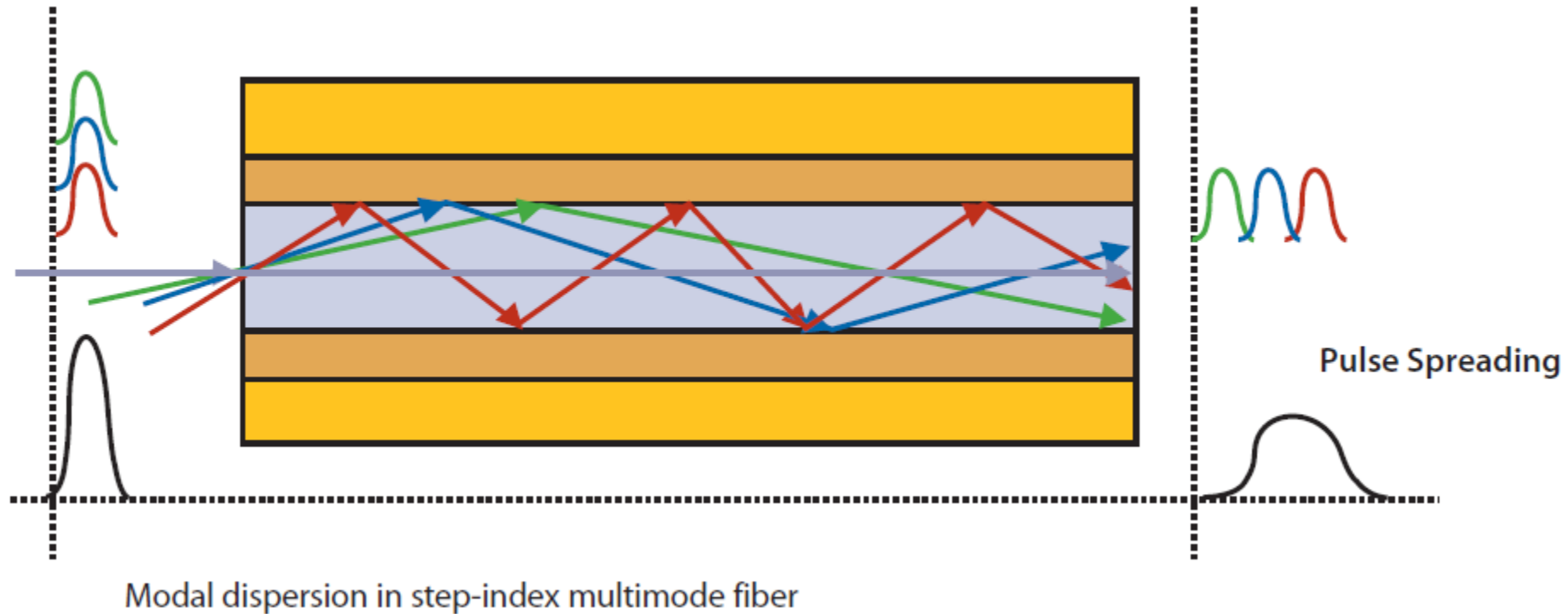
Plastic Optical Fibre (POF)

- large core ~1 mm
- usually step index
- short, low speed

PCS, HCS

- PCS plastic clad silica
- HCS hard clad silica
- smaller glass core ~200um
- thin plastic cladding

Fibre Optics – Chromatic Dispersion



CATV Fibre Optics

CATV

Most CATV applications

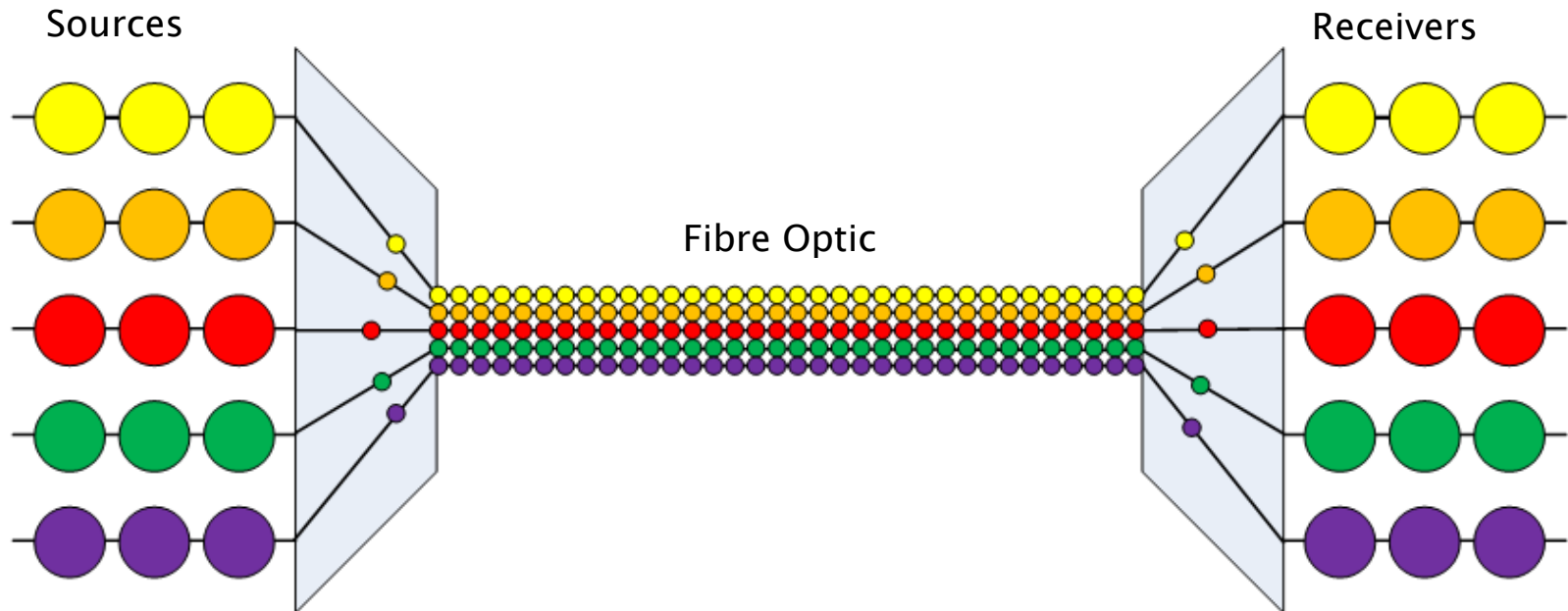
both single mode and multimode signals
within different areas of the network:

- Headend –to– fibre nodes: Single Mode fibre
- then converted to multimode for remaining outward distribution

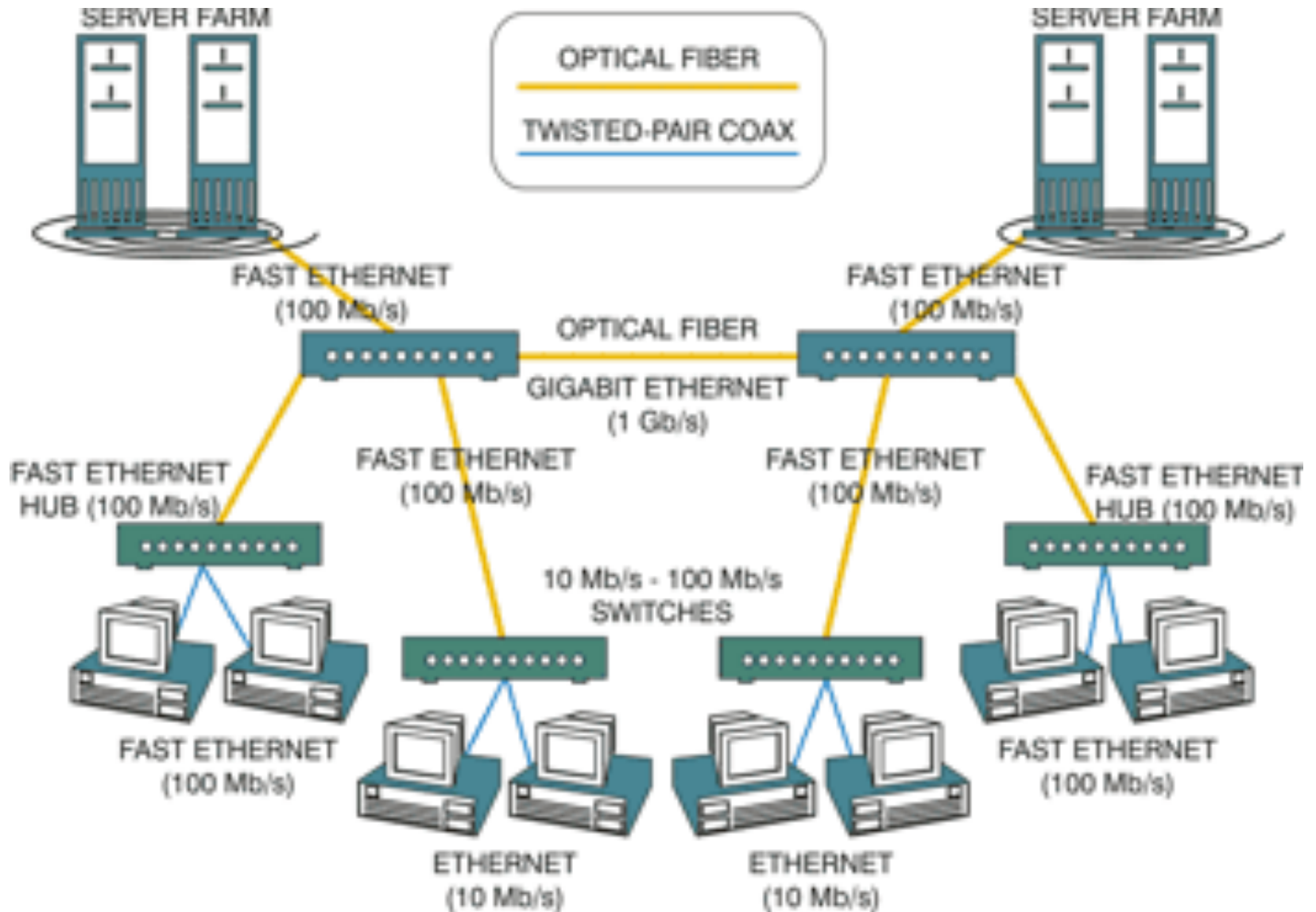
WDM – Wave Division Multiplexing

A number of multiplexes (each of many channels) are carried on one fibre

- Each multiplex is a string of data, each on its own wavelength
- Uses single mode fibre



Fibre Networks



Dark Fibre

- network infrastructure – mixture of cabling, switches and repeaters
- no data = no light: ‘dark’ .. not in use
- Originally – When Networks Planned / Built:
 - usually overestimate cabling required, so many fibres dark, for future expansion
- Plus, now – Dark Fibre often leased from network providers and operators
 - client create / operate their own privately-operated optical fibre network
 - .. rather than just leasing bandwidth
- Could be dark fibre rings, point to point or point-to-multipoint configurations
 - clients aim for high performance, a high security, superfast
 - better signal strength
 - more immune to interference than fibre on traditional networks
- DWDM (Dense Wavelength Division Multiplexing)
 - key in the development of Dark Fibre
 - multiple data signals are transmitted simultaneously, different wavelengths
 - higher capacity

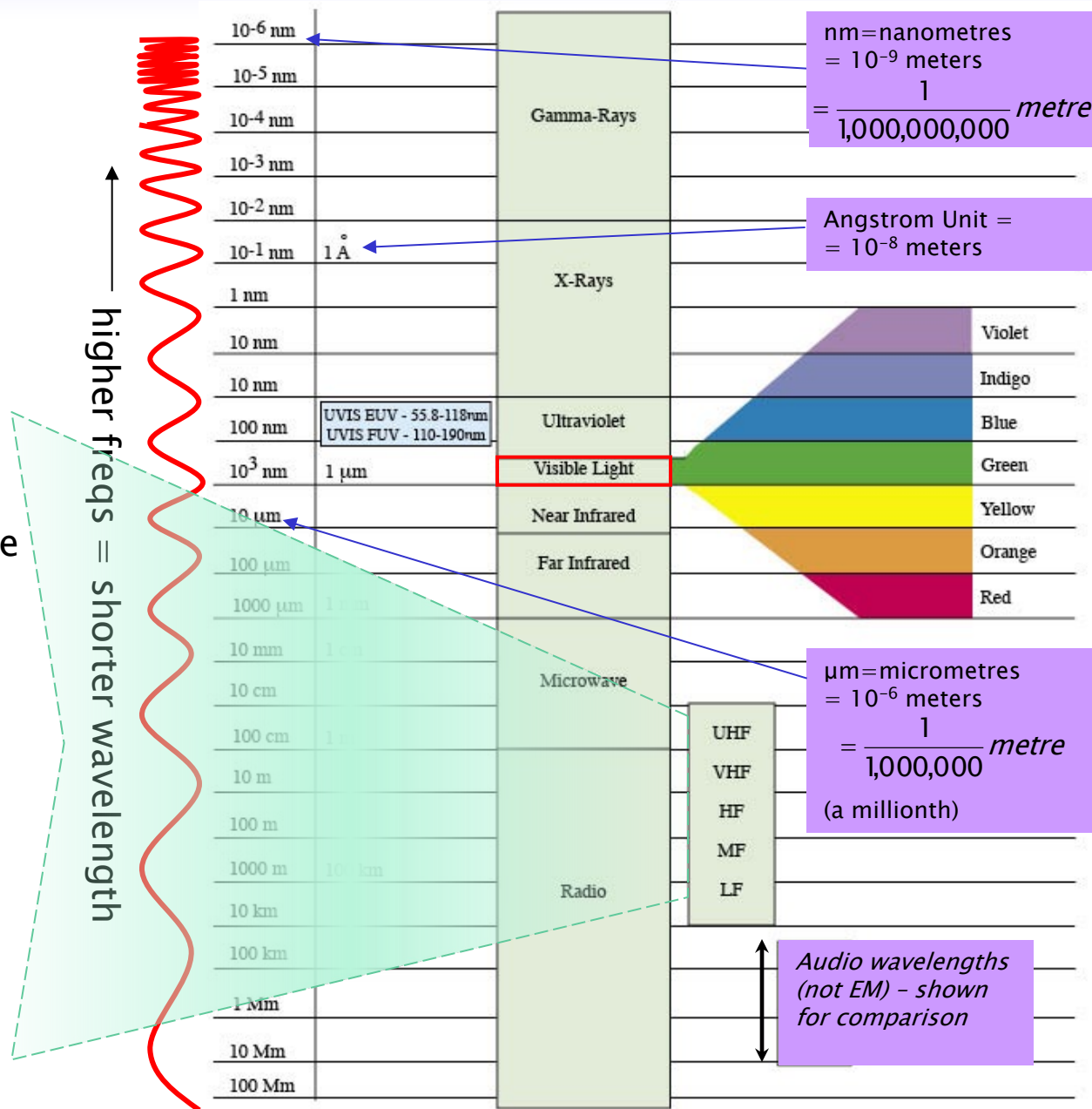
Electromagnetic Spectrum; Radio – LF to VHF

HF and VF Bands:

- Tend to be absorbed by the Earth
- Can be bounced off of the ionosphere (called *sky wave propagation*)

VLF, LF and MF Bands:

- Range of about 1000km
- Penetrate buildings
- Omnidirectional



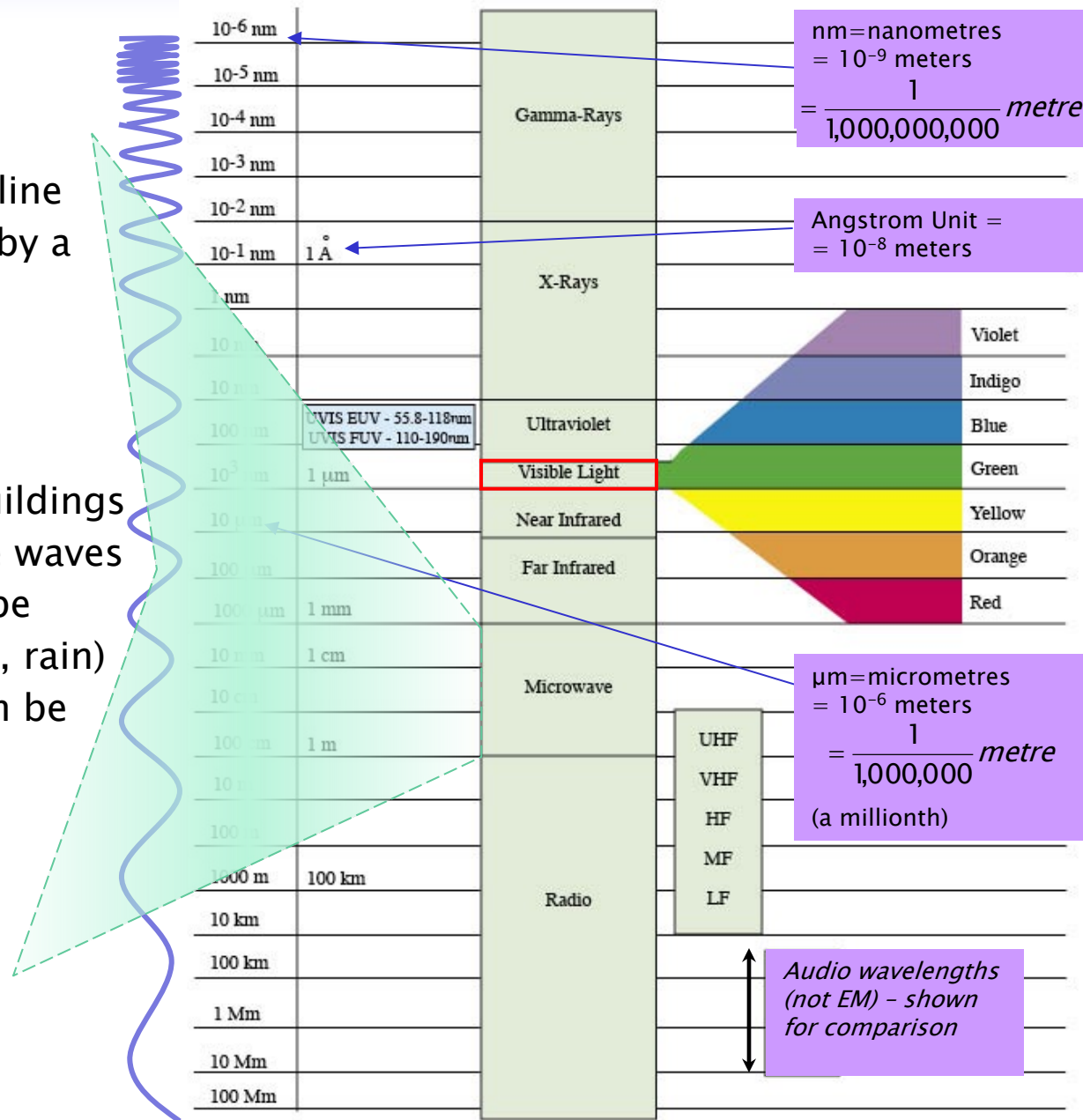
Radio – Microwave

Properties:

- Travel in a nearly straight line
- Can be narrowly focused (by a parabolic antenna)
- High signal to noise ratio
- Little interference
- Low penetration
 - Do not pass through buildings
 - At 4GHz and above, the waves are very short and can be absorbed by water (e.g., rain)
 - At 60GHz the waves can be absorbed by oxygen

Range:

- 2GHz–6GHz: 70km
- 18GHz–45GHz: 1.6–8km

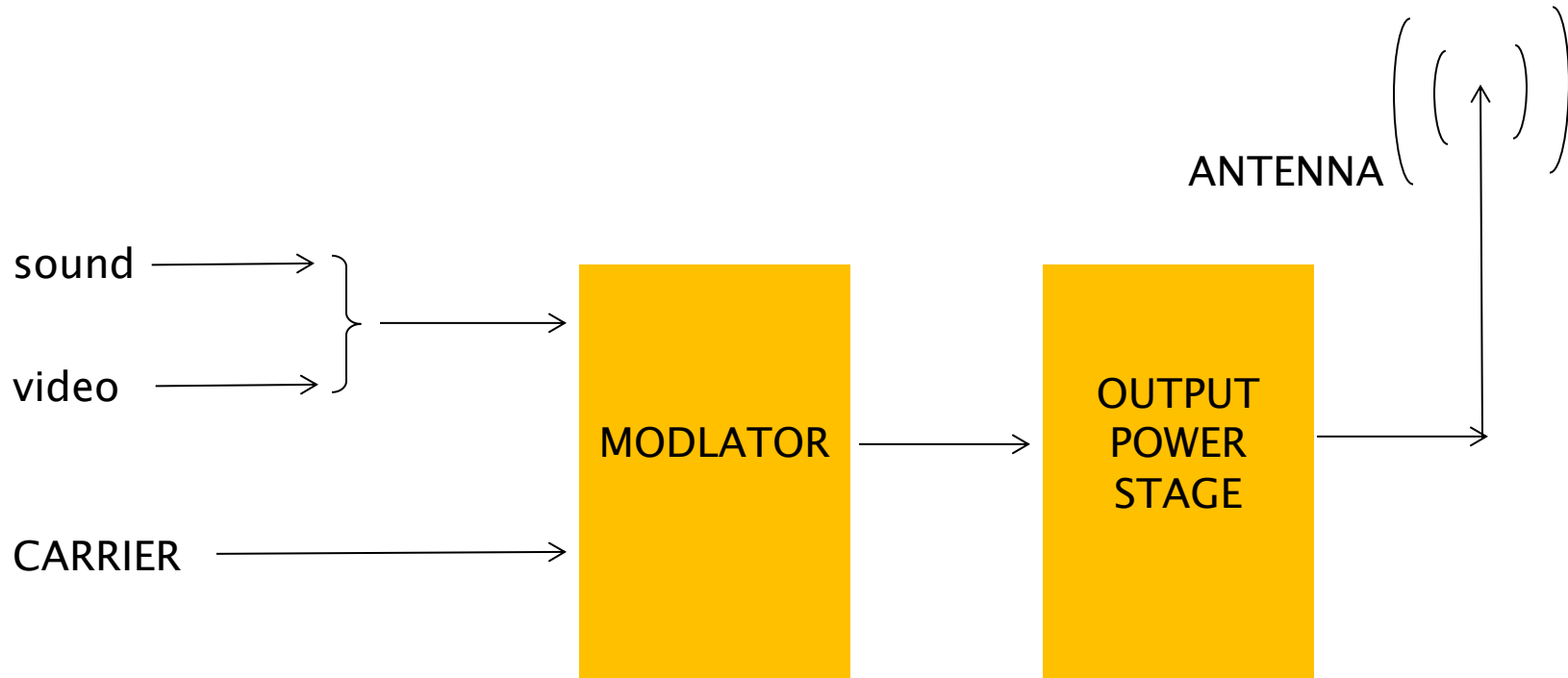


Radio and TV Frequency Band Allocations

<u>BAND</u>	<u>Freqs, MHz</u>	<u>Typical usage</u>
LW	0.14 – 0.28	BBC Radio 4 LW (0.198MHz)
MW	0.58 – 1.6	Medium Wave Radio Stations
HF	3 – 26	special broadcast services, popular for Pirate radio
CB	26 – 28	CB
Low VHF	29.7 – 47	Military, Model aircraft, hospital
Band I	47 – 68	old TN band; now alarms, cordless phones, car alarms
Low Band	68 – 87.5	Military, emergency services, private business radio
FM, Band II	87.5 – 108	FM radio broadcast
Air Band	108 – 137	Aero comms, navigation
Mid Band	137 – 165	Military, police
Marine Band	156 – 163	Marine, ports, navigation, distress
High Band	165 – 174	Medical, telemetry, theatres
Band III	174 – 230	special applications, eg via JFMG freq allocation
NATO Band	230 – 400	NATO, space shuttle
UHF1 & 2	400 – 470	Navigation, repeaters, telemetry
TV, Bands IV & V	470 – 854	TV broadcasting, PAL, SECAM
GSM & mobile	854 – 960	GSM mobile, aero, navigation, astronomy
Amateur 23cm	1.2GHz	Amateur use
Microwaves	1 – 300GHz	Communications links
Ku TV Band	10.7–12.75GHz	Comms, satellite support
Visible Spectrum		

TV/Radio Broadcast Transmitter

- electromagnetic energy from output signal is propagated from the antenna
- may be unidirectional, or focussed in particular direction by antenna design



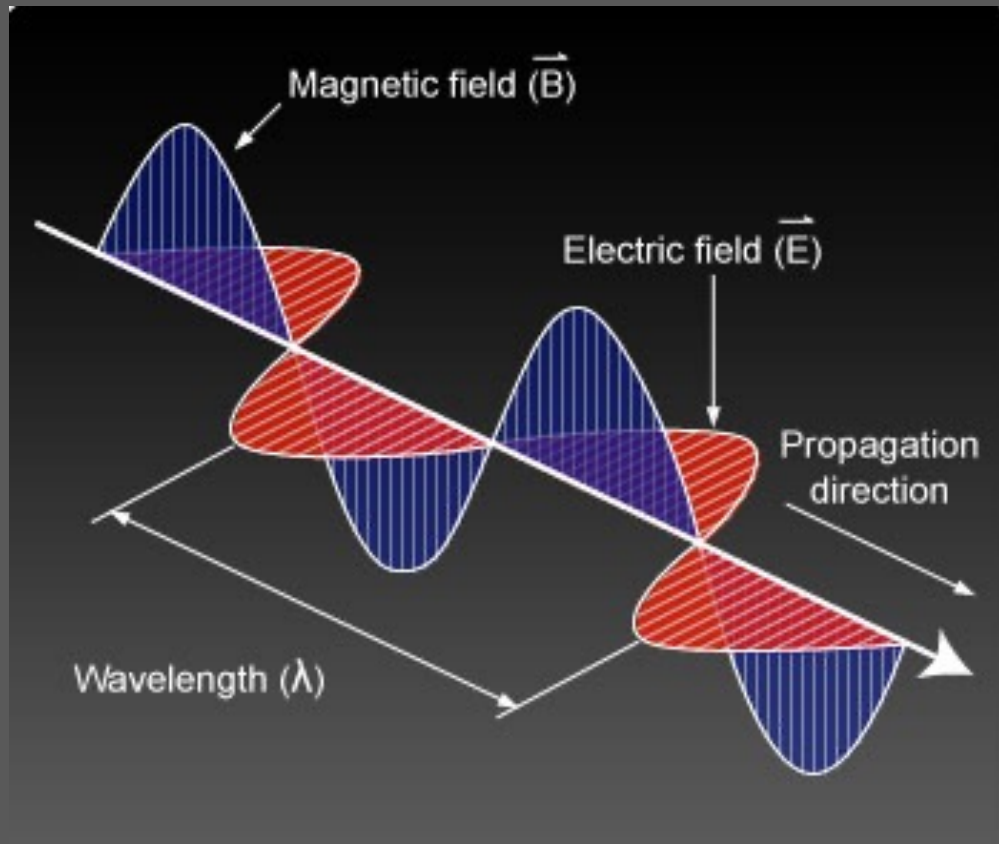
Output stage

- small transmitters – may be solid state
- large transmitters – valves more efficient



**NEC DTU-52 Transmitter
equipment, Winter Hill, UK,
for Arqiva, transmission
provider for UK terrestrial TV**

Electromagnetic Wave

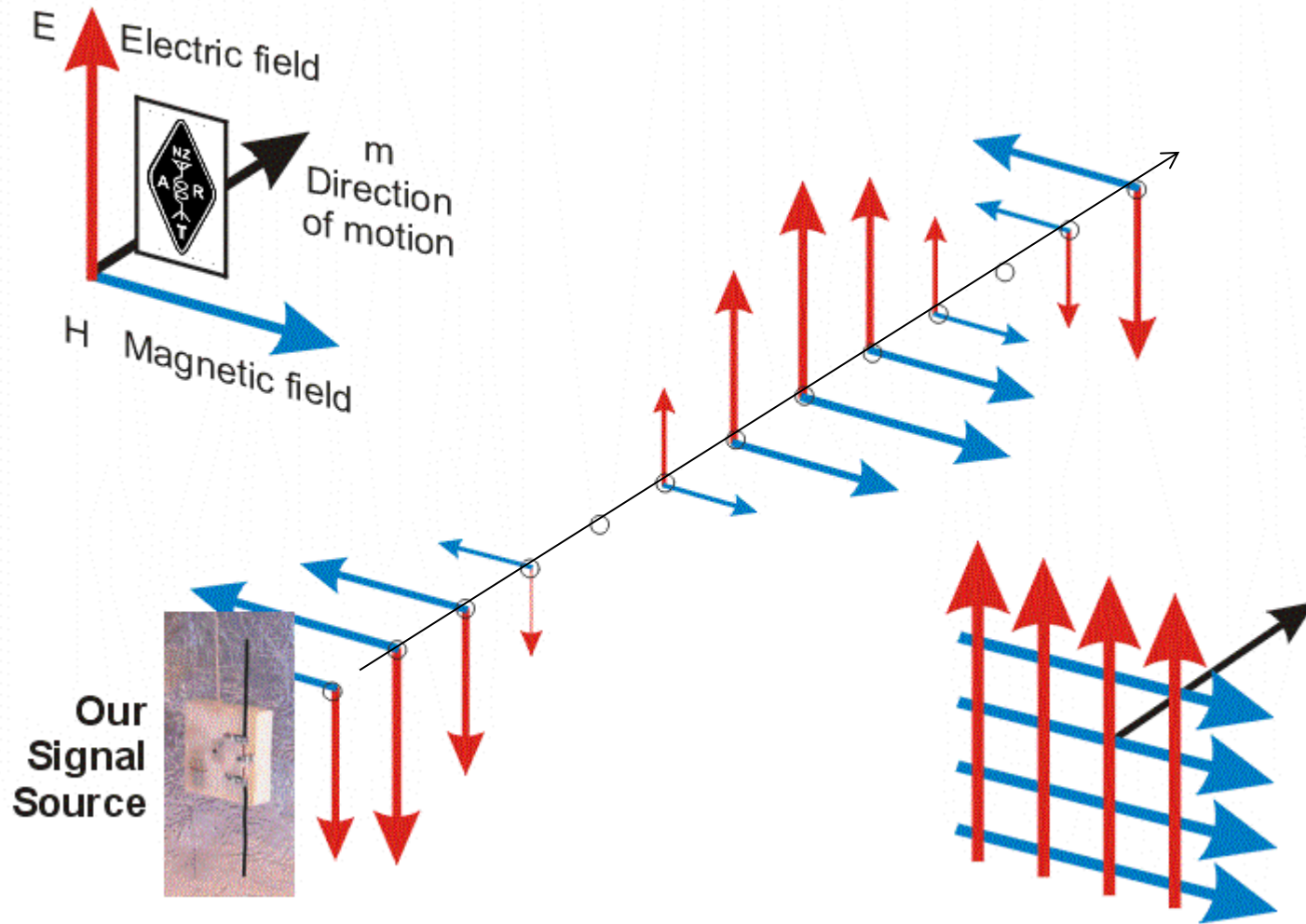


- alternating electrical current produces an electric and magnetic field that are also alternating
- the wave of the electric field and the wave of the magnetic field are propagated perpendicularly to the direction of propagation and to each other

(left)

<http://learn.uci.edu/oo/getOCWPage.php?course=OC0811004&lesson=005&topic=002&page=21>

Electromagnetic Wave



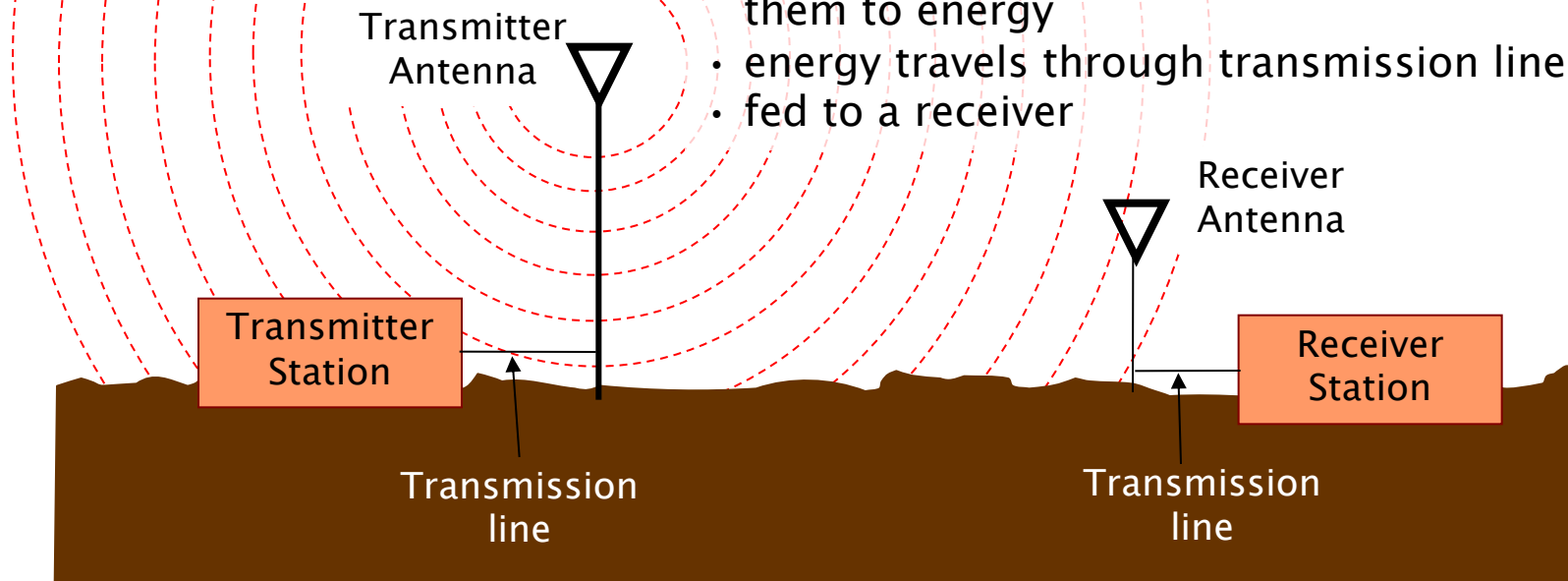
Radio Waves – Basic Antenna

Transmitter – generates radio-frequency energy

- energy travels through a transmission line to an antenna
- antenna converts the energy into radio waves
- radio waves radiate into space from the antenna at the speed of light
- radio waves travel through the atmosphere or space until they are either reflected by an object or absorbed

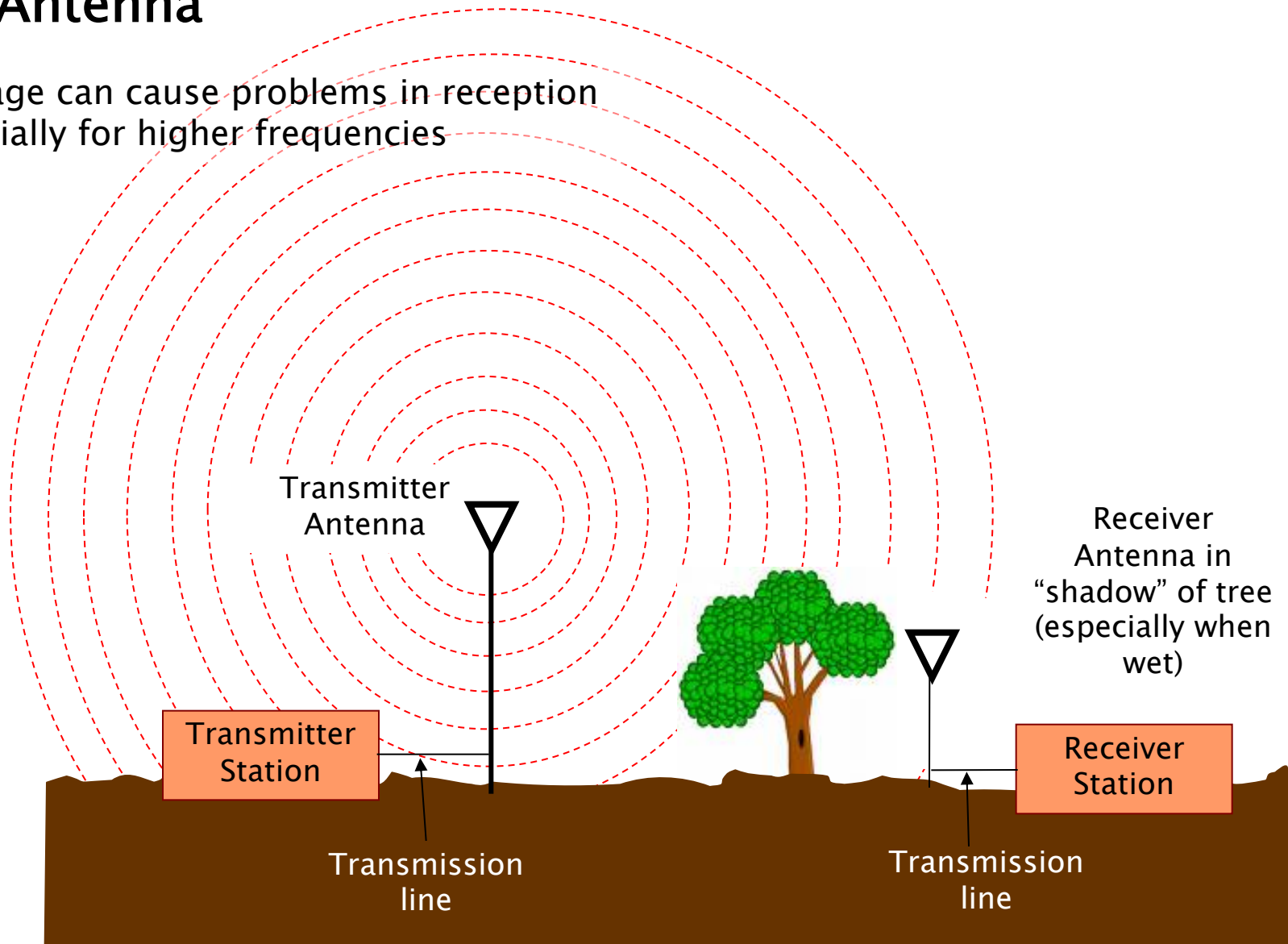
Receiving antenna

- in the path of the radio waves
- absorbs part of the waves and converts them to energy
- energy travels through transmission line
- fed to a receiver

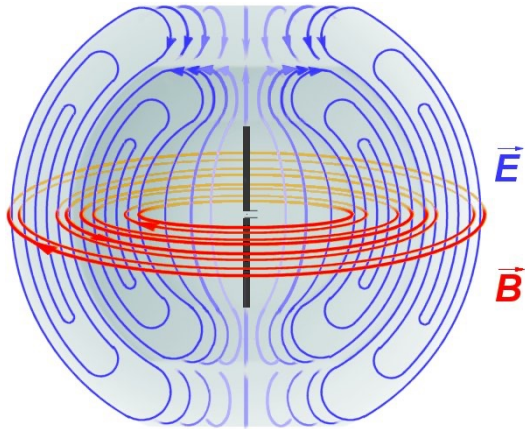


Basic Antenna

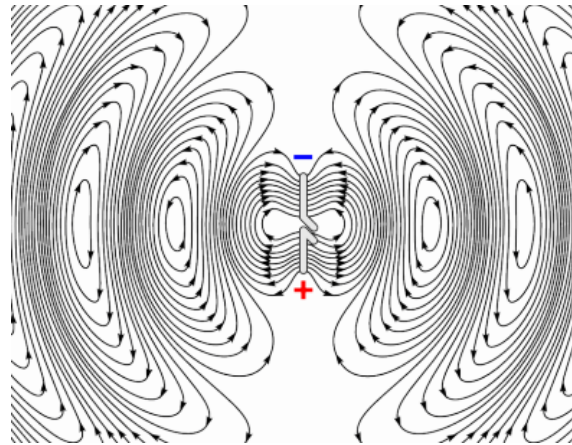
Wet foliage can cause problems in reception
... especially for higher frequencies



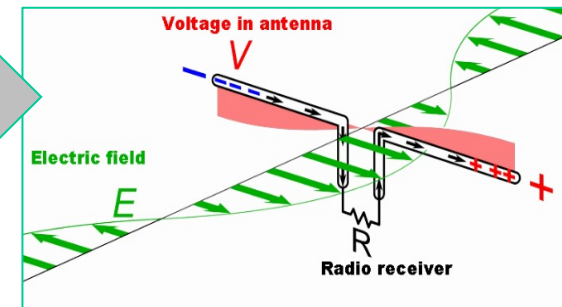
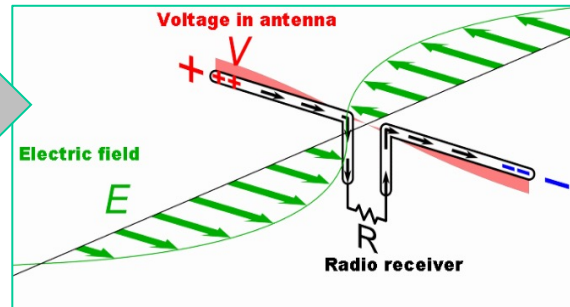
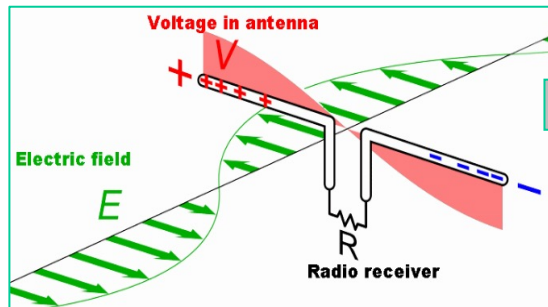
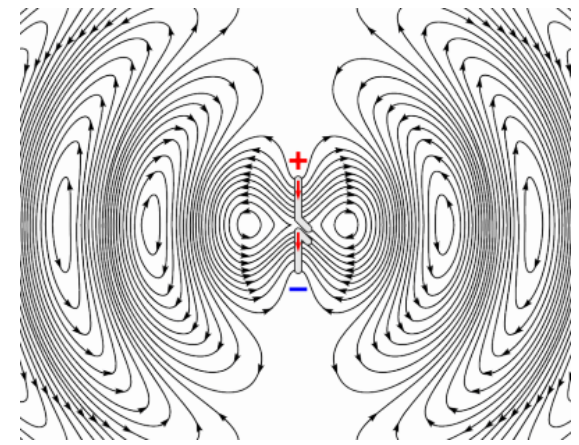
Radio Transmission and Reception



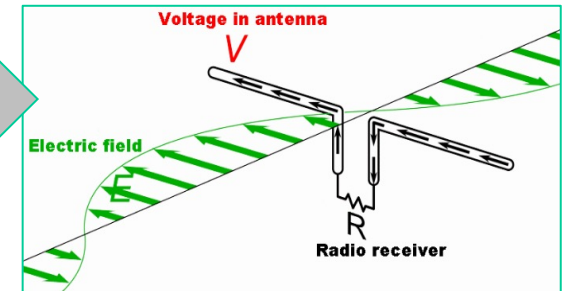
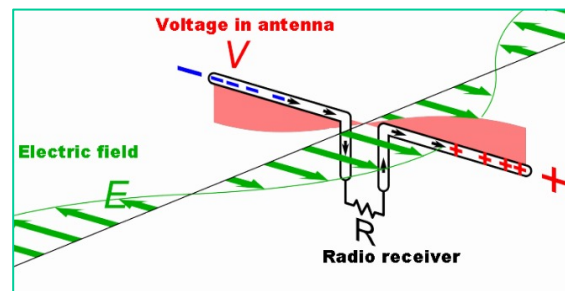
Electric and magnetic fields generated by a dipole antenna



Radiating electrical field from transmitter

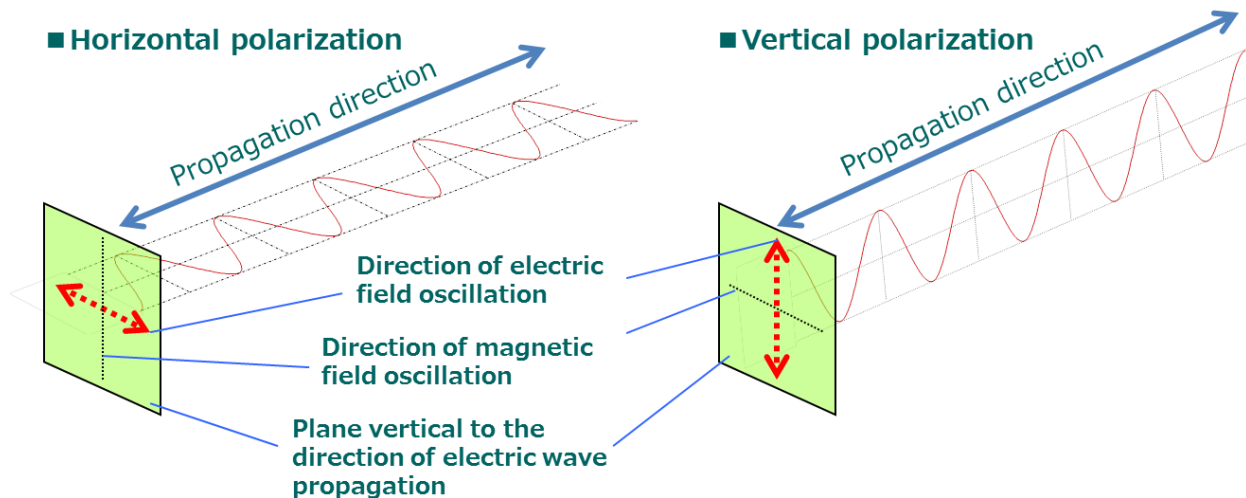


Induced electric and magnetic fields in receiver
Note charge and current directions



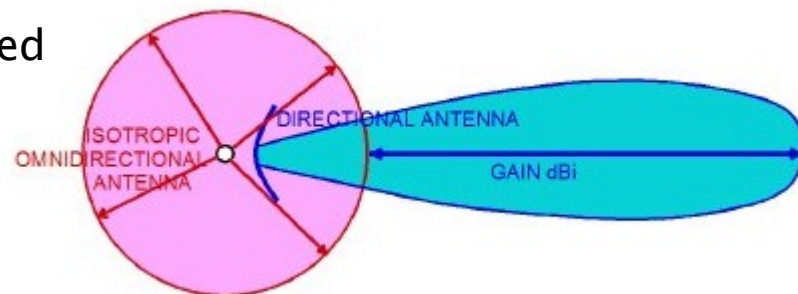
Radio Transmission and Reception

Antenna / transmission polarisation



Antenna Power Gain

- combination of directivity and electrical efficiency
- Transmitting antenna – how well the antenna converts input power into radio waves headed in a specified direction
- Receiving antenna – how well the antenna converts radio waves arriving from a specified direction into electrical power



Radio Wave Propagation

For reception, antenna length at least $\lambda/4$

Low freq (e.g. AM services)

- earth ground has strong influence
 - induce current in ground
 - attenuated by absorption in earth
 - attenuation increases with increase in freq
 - maximum range depends on ...
transmitted power, frequency (< few MHz)



Surface Waves

Higher Frequencies – VHF, UHF

Waves can be refracted off Troposphere

(lowest of atmospheric layers)

refraction due to atmospheric refractive indices

(so not a guaranteed performance due to changing weather)

... see later ...

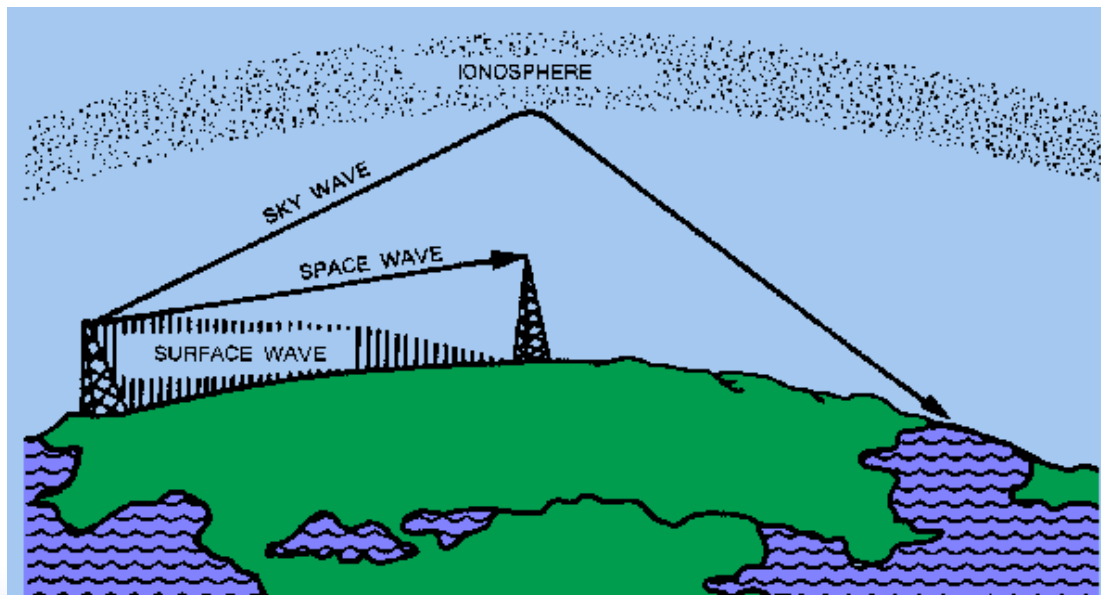
Radio Wave Propagation

Ground waves

- propagate along the earth surface, close to the ground
 - (never reach the ionosphere)
- typically travel 150Km in daytime
- higher freqs are subject to a high attenuation
 - ... travel less than 15 km (at 30 MHz)
- mainly for low frequencies below 1.8 MHz (MW, LW and VLF)

Space waves / Line of Sight

- directly from antenna to another without reflection on the ground
- also assisted by waves bending near the ground (curved path on earth surface)
- Sky wave propagation:
next slide ...



Radio Wave Propagation

Troposphere

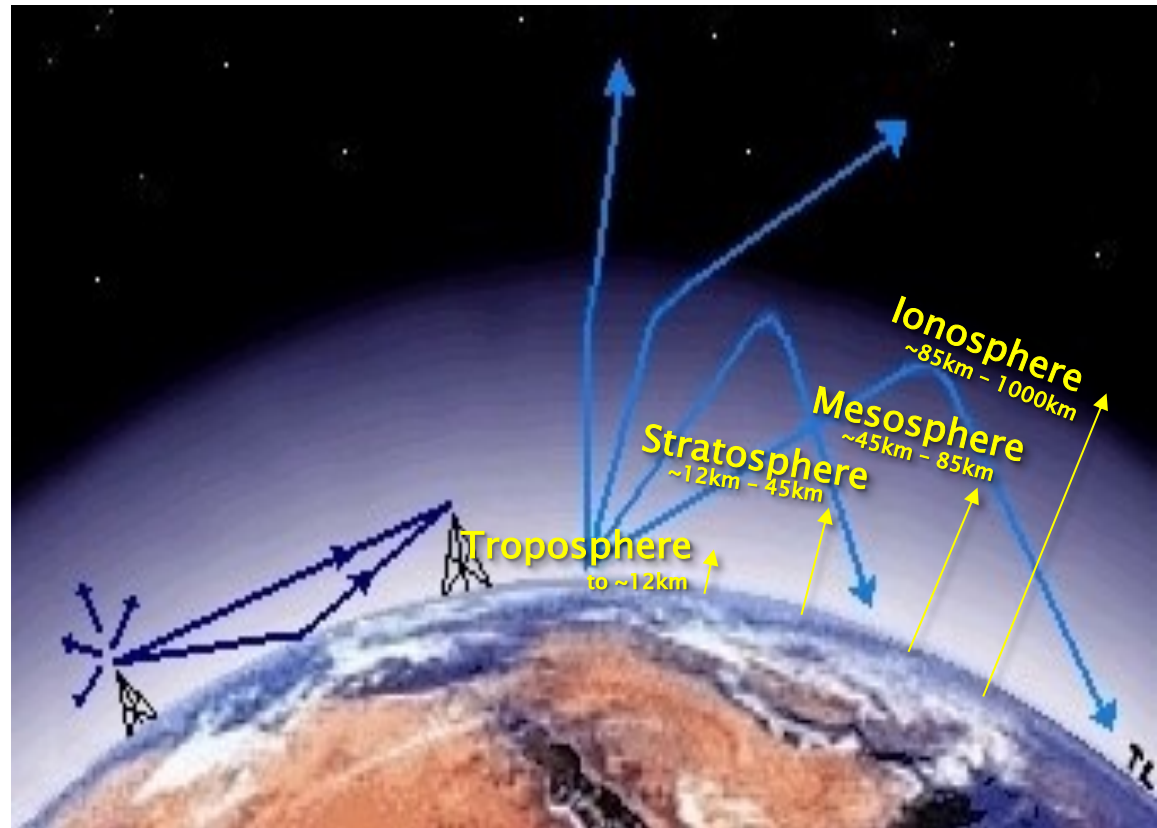
- where all weather takes place; rising and falling packets of air
- air pressure at the top of the troposphere is only 10% of that at sea level (0.1 atmospheres)

Stratosphere

- air flow is mostly horizontal
- thin **ozone layer** in the upper stratosphere
 - high concentration of ozone
 - warm: absorbs UV radiation from the Sun

Ionosphere

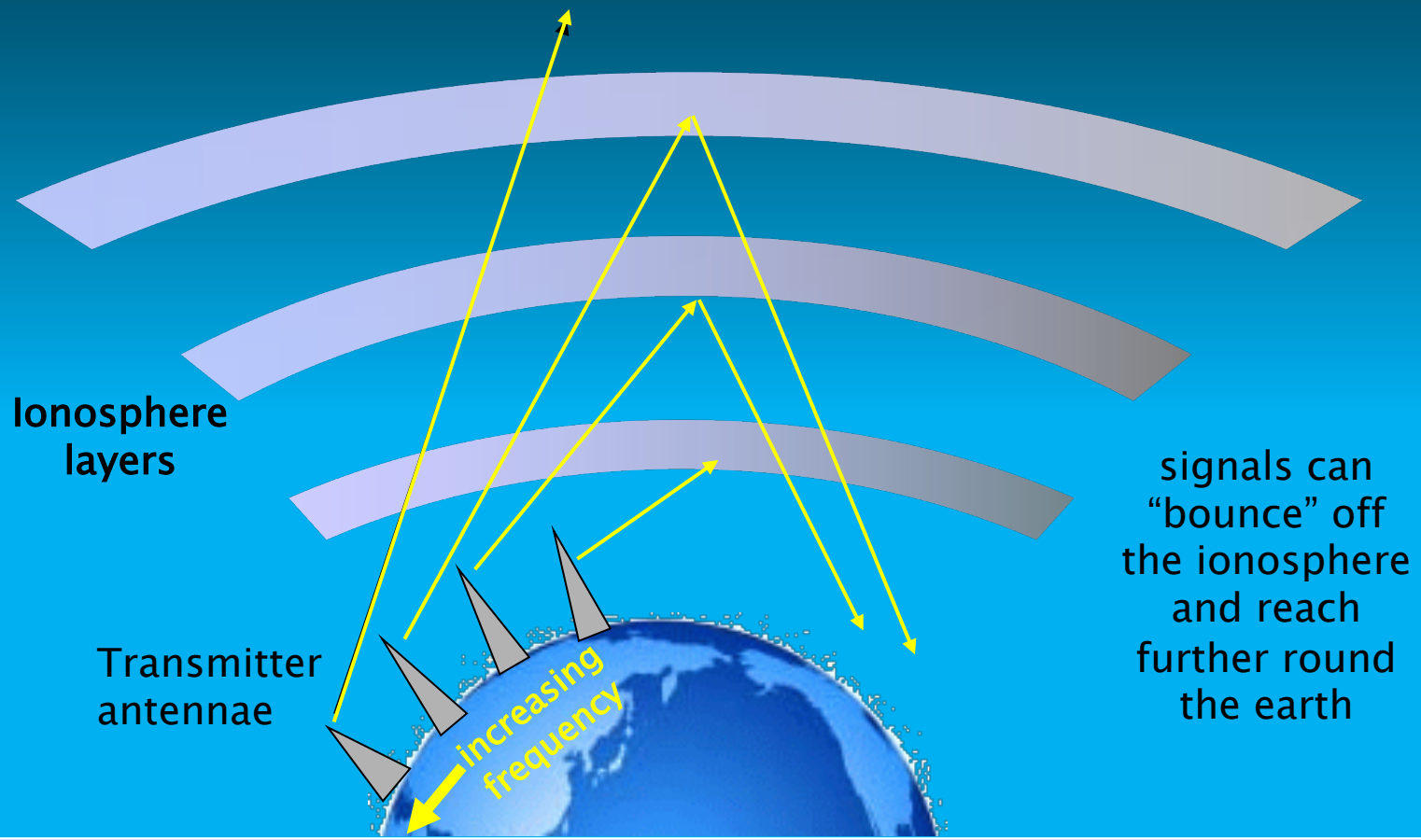
- (generally) ionized by solar radiation ($\therefore$ aurora)



Radio Wave Propagation

Refraction

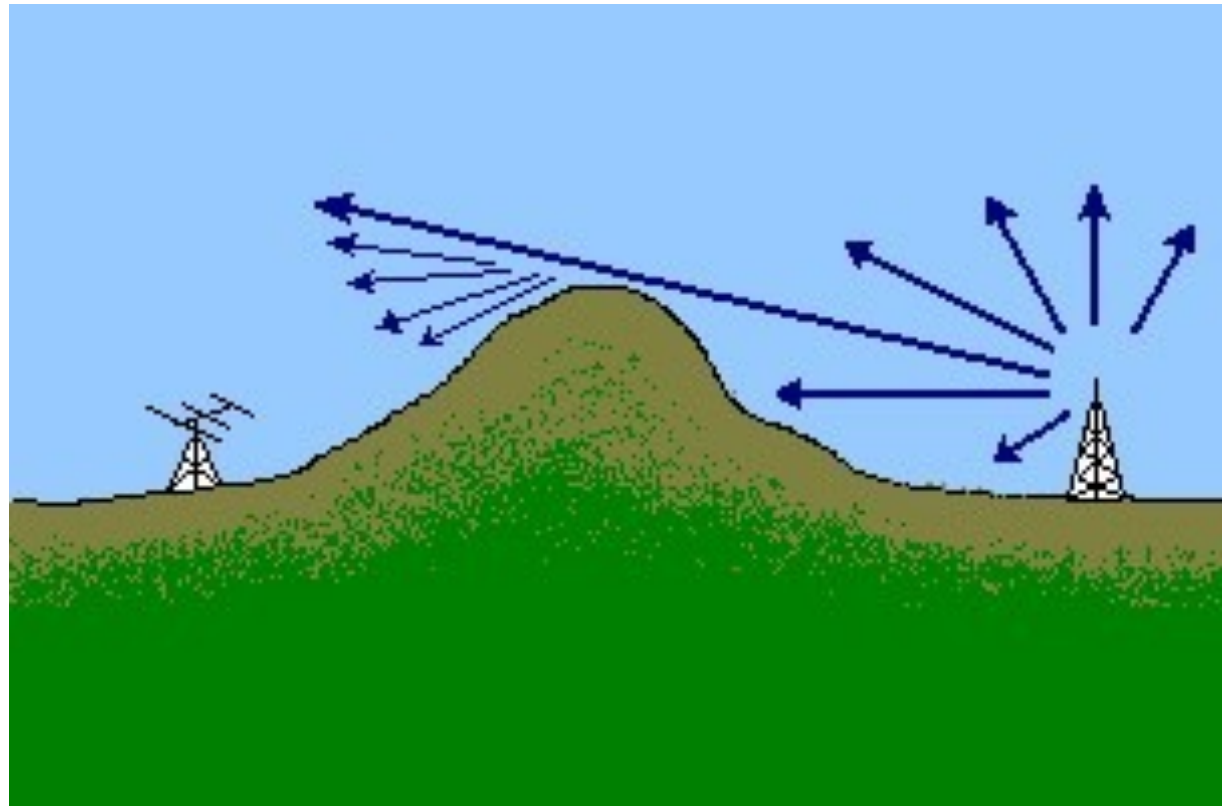
- bending of waves passing through different media (air or ionosphere)
- produces variation in the velocity of waves
- Ionospheric scatter
 - Ionosphere – ~80Km – 600Km+ above the earth



Radio Wave Propagation – Localised Issues

Diffraction

- location is out of sight of transmitter (e.g. – behind a hill)
 - can receive transmissions (but reduced)
- (although HF will more likely reach due to refraction or reflection in the upper atmosphere)



Radio Wave Propagation

Reflection

- similar to light

for long wavelengths (80m+)

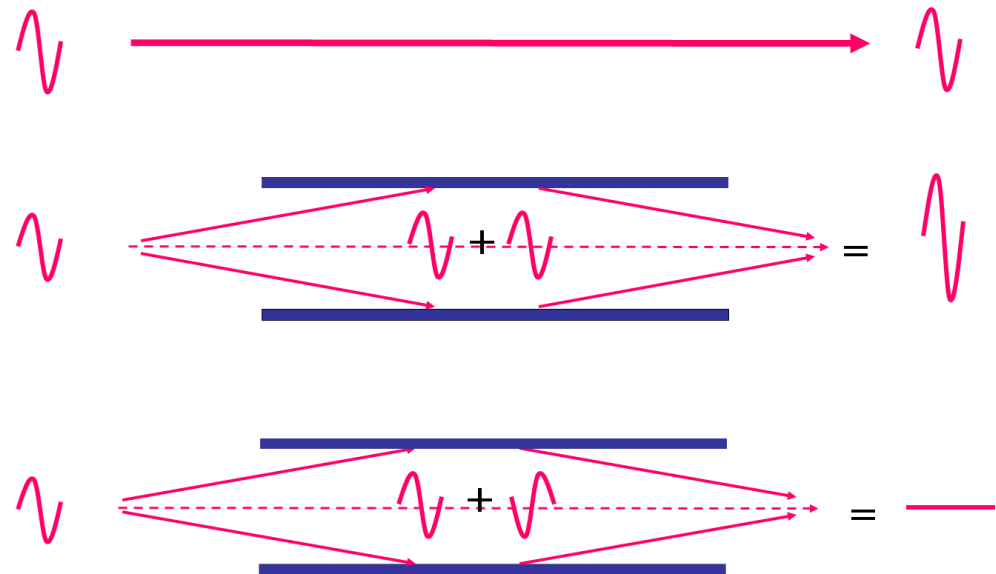
- small obstacles (cars, trees) have little effect
 - objects small compared to the wavelength, can't reflect the energy
 - long waves are reflected by the ground ... but also penetrate

short wavelengths – V/UHF waves (2m – 70cm)

- very sensitive to small obstacles
- metal objects can be used as reflectors

Reflections

- e.g. wi-fi in metal framed buildings
- radio waves reflect and interact
- could cause stronger or weaker reception

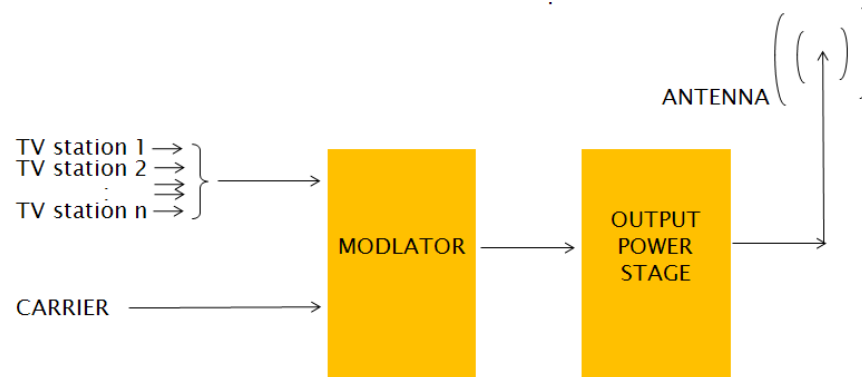


Broadcast Output Power Stages

Years ago – tetrode and pentode valves mainly used on V.H.F. and U.H.F.

- Over years, broadcast on higher frequencies – dimensions of the electrodes decreased in order to reduce their effect on the tuned circuits
∴ reduction in power

- Magnetron developed, then klystron



- Now – more efficient Inductive Output Tubes (IOT) or Klystrons (right)
- capable of producing power output up to about 30 kW continuous and 7 MW pulsed and gains of 20 – 23 dB at frequencies up to about a GHz



<http://www.theonlineengineer.org/TheOLEBLOG/radio-waves-a-tutorial/>
<http://www.thevalvepage.com/valvetek/umvalve/umvalve.htm>
http://www.lbl.gov/MicroWorlds/ALSTool/ALS_Components/RFSysTem/
<http://steveblank.com/category/secret-history-of-silicon-valley/page/2/>
<http://www.theonlineengineer.org/TheOLEBLOG/how-iots-work-a-tutorial-2/>
http://www.youtube.com/watch?feature=player_embedded&v=ecxKfZ3ATwg

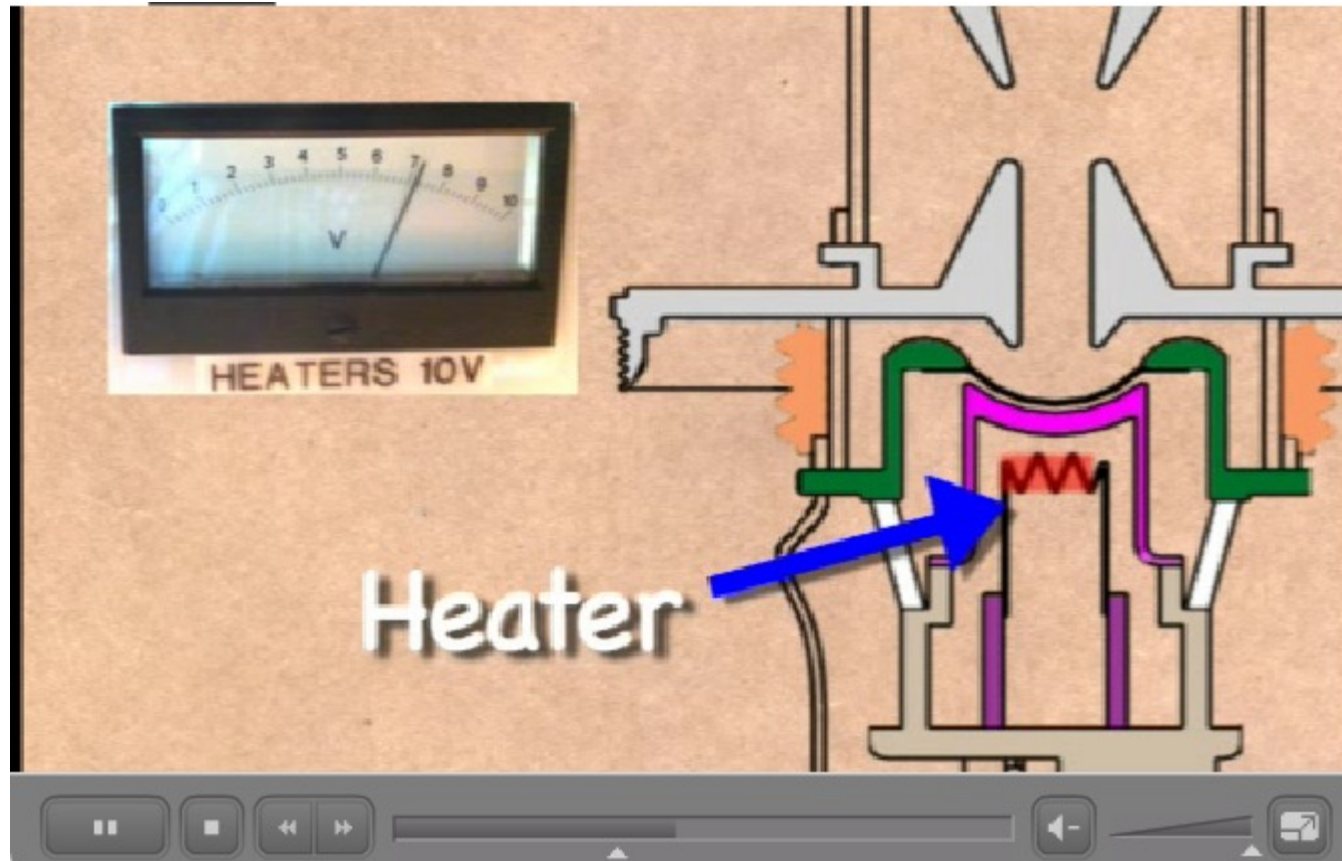
Broadcast Headend Transmitter

IOTs – High power devices

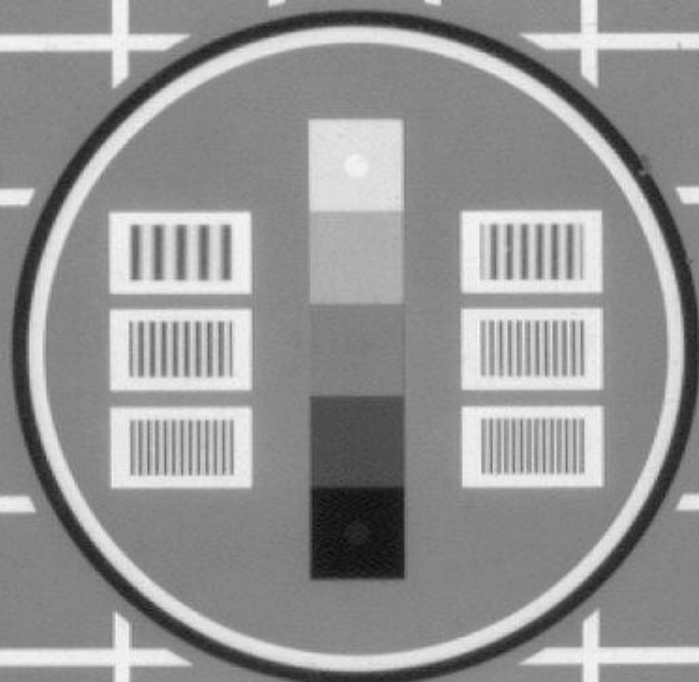
- Need special management for cooling
- Air Conditioned environment
- Water-cooled circuits
- Safety features



How IOTs Work



<https://www.youtube.com/watch?v=fx0asm1kvRw>



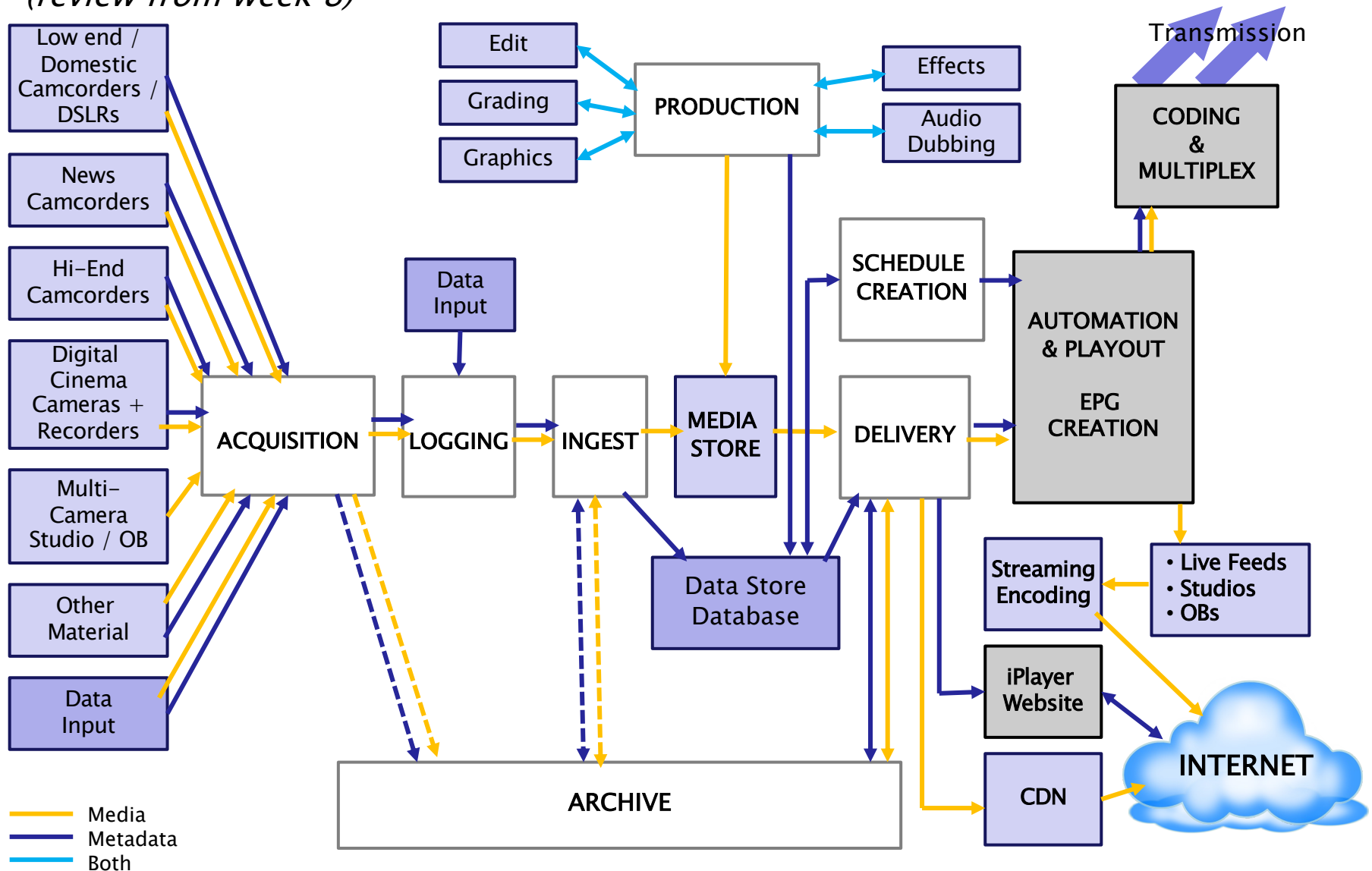
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BBC 1 405 LINES

The Signal Path

Broadcast and Production Workflow (BBC Schematic)

(review from week 8)



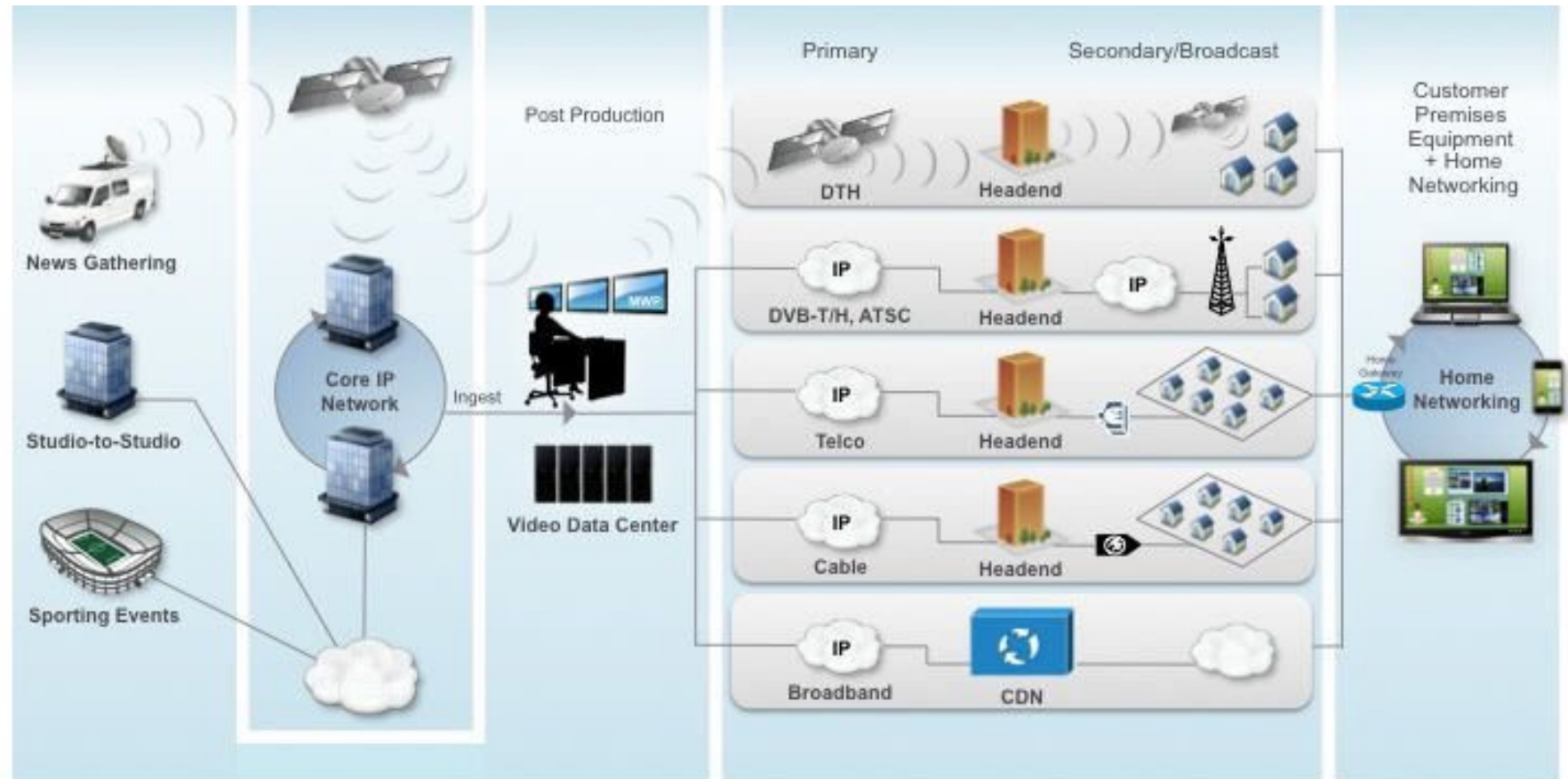
Contribution and Distribution

Production

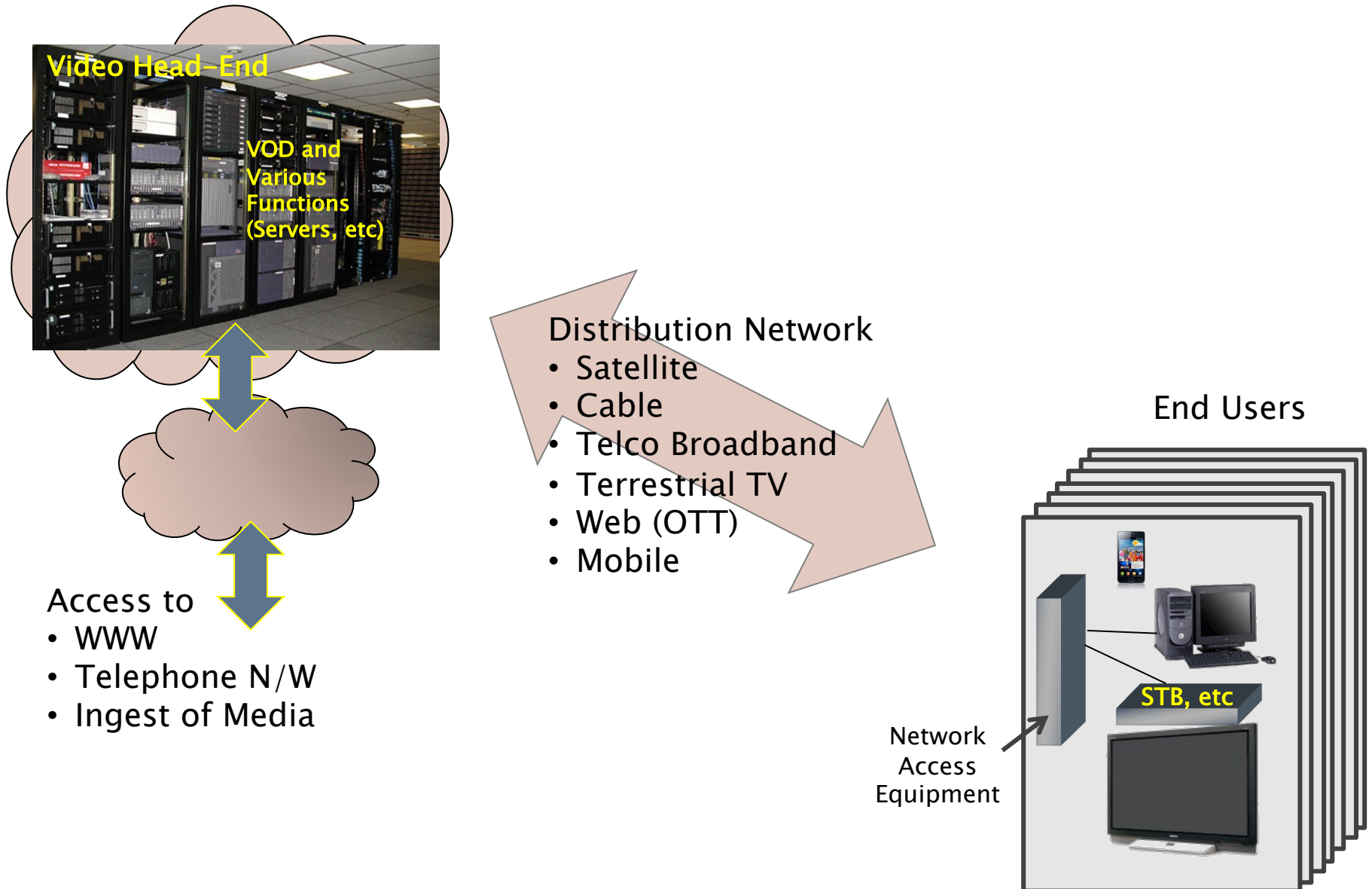
Contribution

Distribution

Consumption

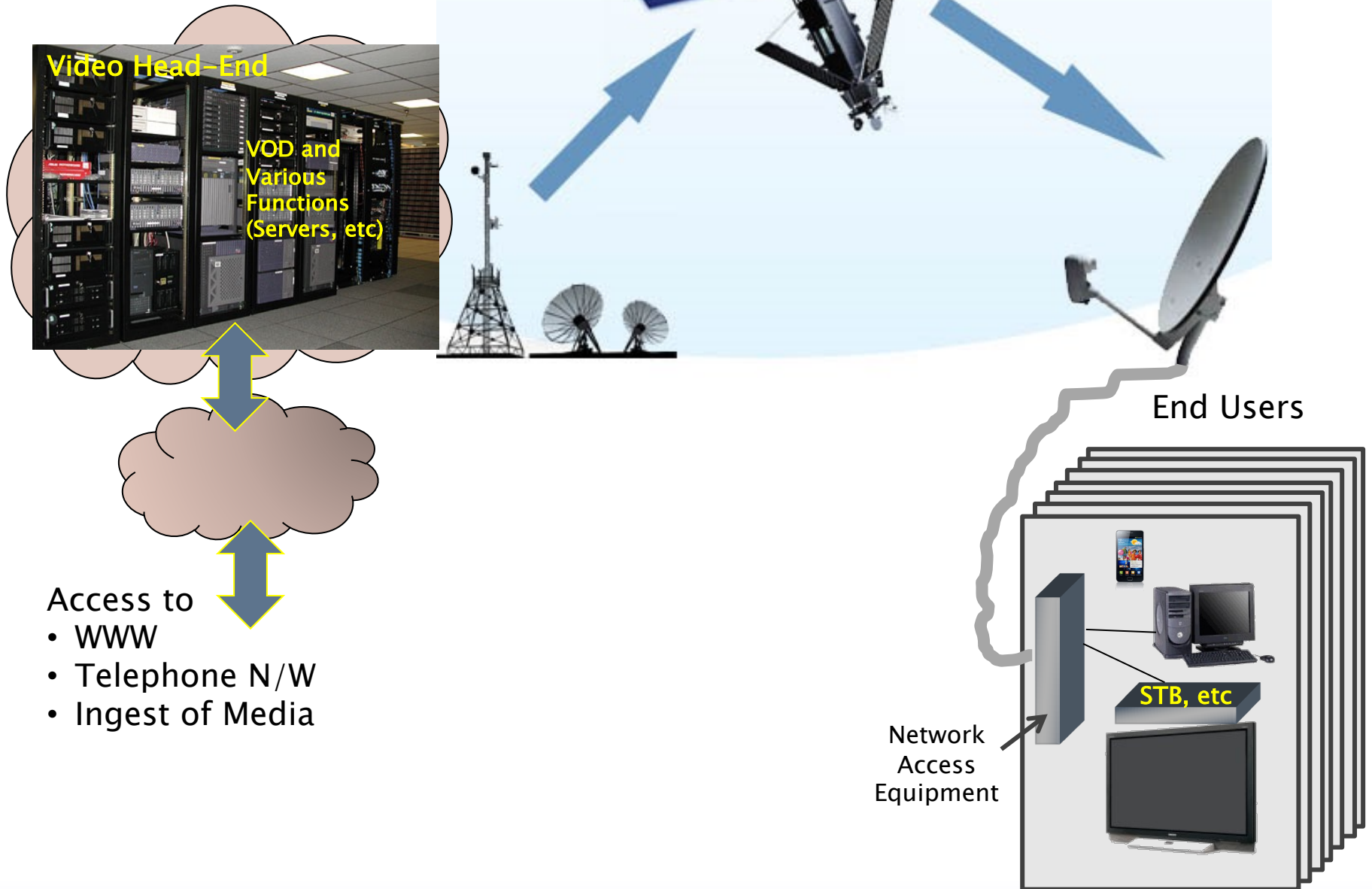


Media Distribution

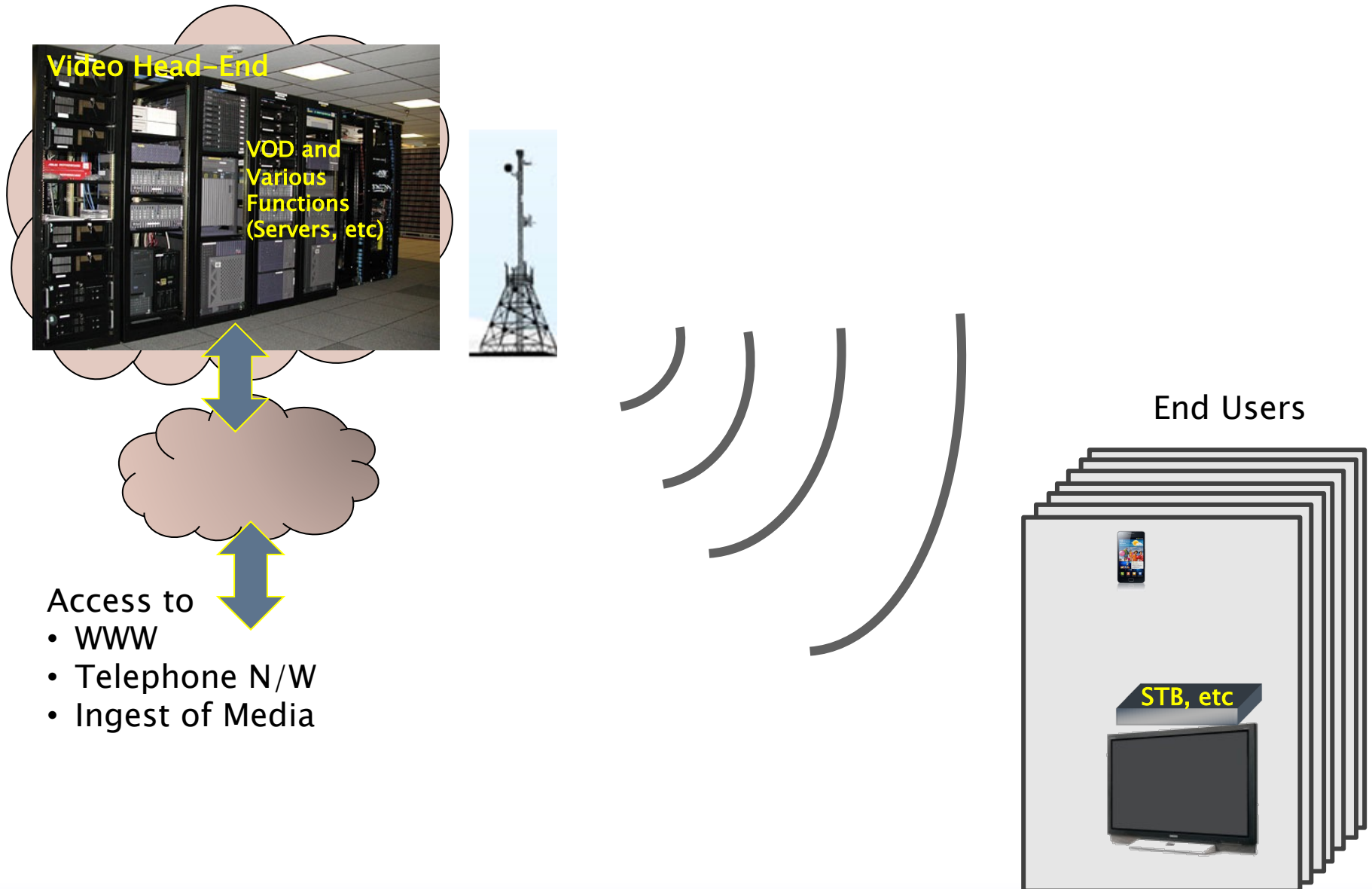


Media Distribution ...

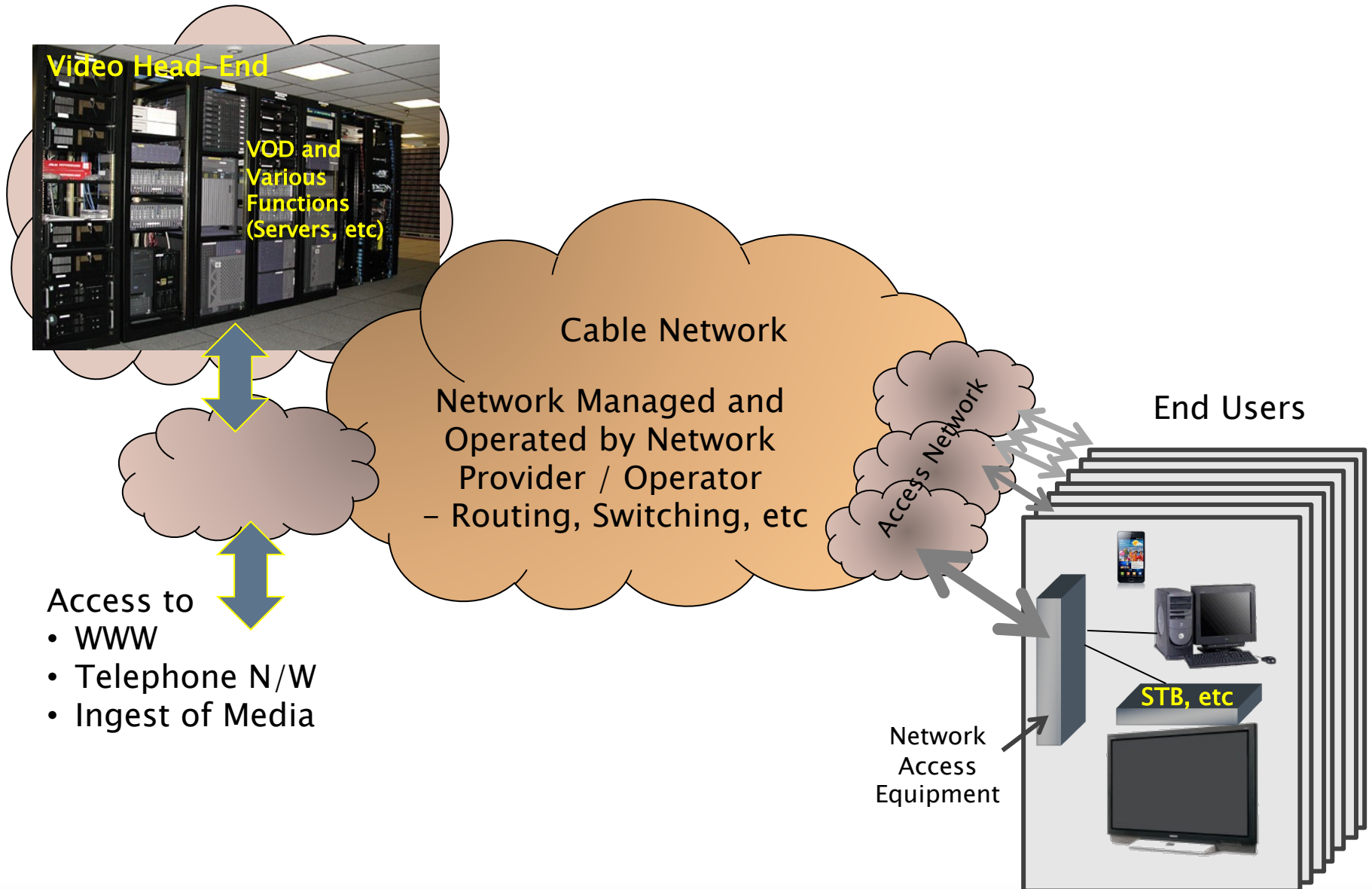
... Satellite



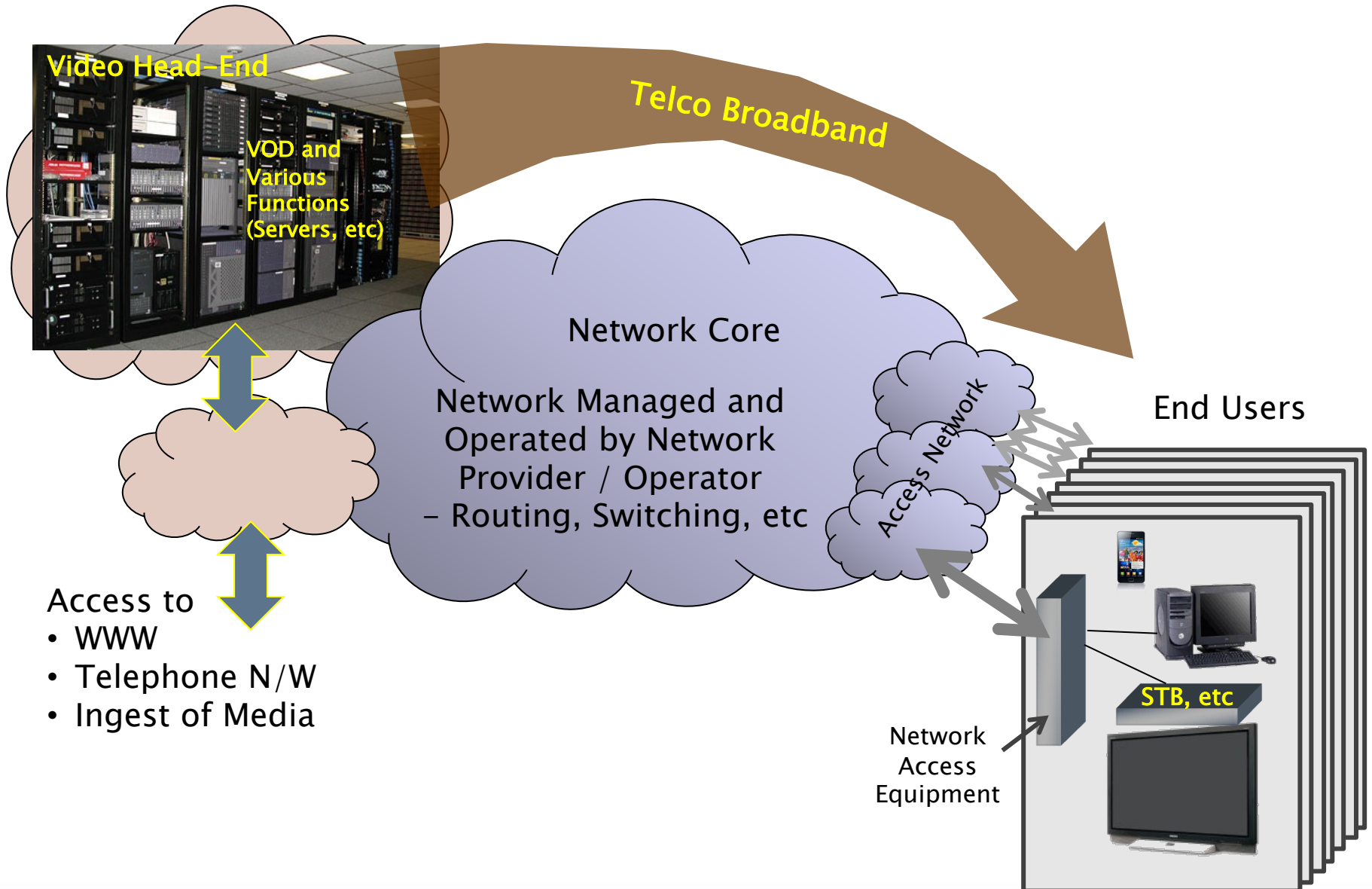
Media Distribution – Digital Terrestrial (DTTV)



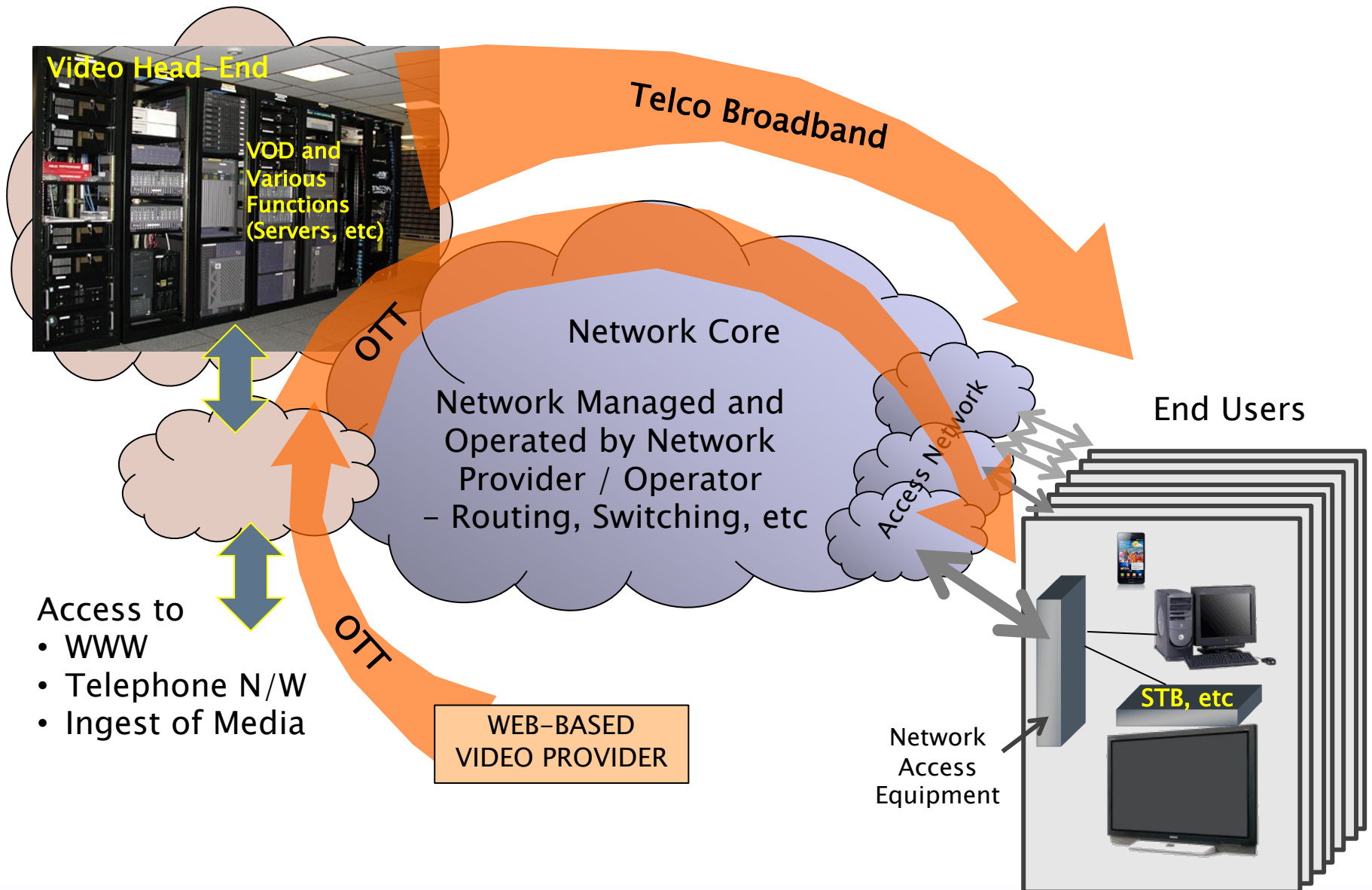
Media Distribution – Cable



Media Distribution – IPTV – Telco Broadband, OTT/Web



Media Distribution – IPTV – Telco Broadband, OTT/Web





F
BBC 1

VHF



UHF



PULL-UP
VOLUME

VHF UHF

CONTRAST

BRIGHT

COLOUR